Practice Questions
STAT 1123, 1124 & 1181 & BUS 232

Table Of Contents

Contents

Table Of Contents 1
Graphs & Description (For Midterm 1) 1
Descriptive Statistics (For Midterm 1) 4
Normal Distribution (For Midterm 1) 7
Chi-Squared Test & Related Probability (For Midterm 1) 11
Probability (Midterm 1) 18
Confidence Intervals (Midterm 2) 20
Regression (Midterm 2) 24
Discrete Probability Distributions (Midterm 2) 43
Sampling (Midterm 2) 49
Binomial Events & Binomial Probability (Midterm 2) 50
Hypothesis Testing (Final Exam) 54
Producing Data (Sampling) (Midterm 2) 55
Tables & Formulas 56

Graphs & Description (For Midterm 1)

1. In each of the following situations, name one graph and one statistic (numerical summary such as mean, quartiles, etc.) that we can use to summarize the data.

a) A track team with 31 athletes keeps a record of running times (in minutes) for runners each day who train for the 3,000 m. Suppose it is known that the distribution of running times is skewed.

b) For a sample of 55 customers, Starbucks records the number of minutes that each customer has to wait in a line-up for latte's.

c) One hundred students are Simon Fraser University were asked in a survey how much they spend (in $) last month on school supplies.

d) Netflix keeps track the genre (action, drama, documentary, or other) of the movies streamed by subscribers. They decided to sample 3000 Vancouverites who were subscribed.

e) A random sample of 100 travelers was drawn to investigate which location in BC – either Grouse Mountain or Capilano Suspension Bridge - they want to see the most.

f) A marketing consultant observed 50 random shoppers at a supermarket. One variable of interest was how much each shoppers spent in the store during the visit. The amount of spending is known to be right-skewed.

g) A statistician wants to understand the relationship between a company's annual revenue (in dollars) and its advertising (in dollars).
h) A sample of 50 employees in a large corporation will be taken to investigate the relationship between the amount spent dining out per month (< $200, $200-$400, or >$400), and marital status. The employees in the sample will be asked to provide the amount each spent dining out per month and his or her marital status (married, single or divorced).
i) As part of a marketing study, 100 lower mainland residents were asked which TV station they regularly check online for local news.
j) To estimate a restaurant's mean sales per day, an audit will analyse the daily receipts for 100 randomly selected days of operation.
k) A store surveyed approximately 100 shoppers in order to investigate whether the frequency of visits to the store (never before, 1-3 times, 4-5 times, more than 5 times) was related to the age of its shopping customers (<= 25, 26-40, 41-55, Over 55).
l) A study was carried out to investigate the relationship between pet ownership and stress. To do this a random sample of 100 people are taken. From each person in the sample was recorded whether they own a pet (yes/no), and their self rating of stress level (on a scale of 1=lowest to 10=highest)
m) A realtor wants to predict the monthly rent for an apartment (n $) based on its size (n square metres). To do this she takes a sample of 25 apartments, ranging in size from 60 to 180 square metres.
n) A company surveys 10 retail chains in order to understand the relationship between sales (in millions of units) of a consumer electronic device and its price (in dollars).
o) A consultant wants to look at the relationship between the number of days a tourist stays in Vancouver and the amount of dollars that they spent
p) A study on telecommuting classifies a sample of students according to year at Langara (classified as 1, 2, 3, or 4+) and the number of study days spent at the school (classified: as none, one, two, three, or more than 3)
q) A climate scientists takes measurements of ocean water temperature (in degrees Celsius) in the Pacific ocean over several locations.
r) A group of statistics students wanted to know if there is any relationship between the daily amount of time (in minutes) on social media students spent and their commute time (in minutes) to school. A random sample of 100 students was drawn in the last week in January.
s) A football fan wanted to see how fast average office clerks could throw a football. A random sample of 50 office clerks in downtown area. It is believed that the throwing speed is skewed.
t) The Vancouver Airport Authority would like to investigate the linear relationship between the number of passengers in line at any given time (numeric) and the wait time per passenger (in minutes). The airport authority randomly selected 200 passengers in the airport to include in their sample.
u) An airline keeps track of the price of tickets (in Canadian dollars) for travellers travelling to a destination. Suppose that the airline takes a sample of 10,000 tickets and records the price of each ticket. The price of each ticket it highly left skewed.
v) A cruise line is looking to understand the variability in the amount of dollars that customers spend on a cruise ship during a cruise. The cruise ship decides to sample 100 cruise travellers on a particular ship sailing. Assume that the data of dollars spent (in CND) is symmetrical.
w) A study to investigate if the distribution of student enrolment across various faculties (natural sciences,social sciences, or humanities) differs between females & males.
x) A study to investigate the relationship between average precipitation and average temperatures for North American cities.
y) A study to test if the average age of Canadians have changed from 1981, when it was reported to be 34.8 years. A random sample of 100 Canadians was collected.
z) One hundred adult Canadians were asked in a survey how much they spend (in $) last month on herbal supplements and vitamins. The distribution of the data is found to be highly skewed. [2 marks]

aa) An ice cream store keeps a record of weekly sales (in litres of ice cream). Suppose it is known that the distribution of ice cream sales is highly right skewed.
bb) For a sample of customers, a bank records the number of minutes that each has to wait in a line-up.
cc) A marketer wants to investigate if consumers from different geographic regions (West, East, South, or North) prefer different types of automobile (Sedan, Coupe, or SUV).
dd) In order to estimate the average household incomes in Vancouver, a city counsellor will survey 300 households. Each household will be asked to provide the household's total annual income (in dollars ($)) in 2018.
ee) An employer wants to investigate the relationship between his employees' daily commute time to work (in minutes) and their annual salary (in dollars ($).

A not-for-profit organization responsible for promoting the safe use of ski hills in North Vancouver. They surveyed a random sample of 200 skiers and snowboarders to determine the relationship between the amount of time (in minutes) the skier spent skiing on the hills in the North Shore and his or her age (in years).

2. What statistic is most appropriate to measure the relationship between the two variables?
3. What graph is most appropriate to measure the relationship between the two variables?

A market research firm wishes to survey 300 skiers to determine what brand of ski they prefer: Nordica, Rossignol, Atomic, Head, Fischer, etc.

4. What graph is most appropriate to display the data provided?

A ski racing coach wishes to determine how long racers take to get down the hill, as well as key statistics for racers racing on this years World Cup circuit. For the 47 racers who raced at the AUDI FIS Ski World Cup in Banff (Canada), the racers recorded the total time that it took for the racers to complete the race (in seconds).

5. It is known that the data is highly skewed. What statistic is most appropriate to measure the variability of the racers time (in seconds).
6. It is known that the data is highly skewed. What statistic is most appropriate to measure the centrality of the racers time (in seconds).
7. What graph is most appropriate to illustrate the total time that it took for the racers to complete the race (in seconds).

A pizza-loving university student, who is living in the dormitory at Simon Fraser University, wanted to see if *Pizza Hut is* taking a longer time than normal to make their pizza delivery these days, since he remembers that pizza used to be delivered within 30 minutes in the past. He recorded the delivery time for the next 40 orders during the year (in minutes).

8. What statistic is most appropriate to measure the **variability** of the pizza delivery time (in seconds).
9. What statistic is most appropriate to measure the **centrality** of the pizza delivery time (in seconds).
10. What graph is most appropriate to illustrate the total time that it took for the pizza to arrive (in seconds).

A basketball fan wanted to know how scoring (Score; defined as average number of point scored per game) of NBA players would differ by their position (Position; classes are guard, forward, and centre) in 2019-20 NBA season. A sample of 200 NBA players was drawn to investigate this. The variable Score was later manually categorized in to four classes, namely: (A) 0-9 points per game, (B) 10-19, (C) 20-29, and (D) 30 or above.

11. What graph is most appropriate to display the relationship between score and position?

12. What statistic is most appropriate to display the relationship between score and position?

A social worker would like to find out if exercise amount (Exercise; classes are low, medium, high) had an effect on the happiness index (Happiness; indices are whole number from 1 to 5) among students. A sample of 80 students was drawn from around SFU.

13. What graph is most appropriate to display the relationship between score and position?
14. What statistic is most appropriate to display the relationship between score and position?

A marketing consultant observed 40 random shoppers at a ski shop. One variable of interest was how much each shopper spent in the store. The amount of spending is known to be right-skewed.

15. What statistic is most appropriate to measure the **variability** of the amount spent in the store (in dollars).
16. What statistic is most appropriate to measure the **centrality** of the amount spent in store (in dollars).

Descriptive Statistics (For Midterm 1)

17. A researcher was looking to understand the distribution of international exports from various states in the United States. To complete this work, the researcher gathered data from 12 states at the end of 2019, with the total value of international exports being expressed in millions of dollars. The 5 number summary from the data is shown below**.**

Min.	Q_1	Median	Mean	Standard Deviation	Q_3	Max
1.6	3.4	4.65	4.65	1.78	5.4	7.7

a) Draw a boxplot of the data using the information provided. (2 marks)
b) If a state has a z-score of -1.58, what is the value of export dollars (in millions)? (2 marks)
c) Are there any outliers in these data? Justify your answer showing the appropriate calculation. (3 marks)

18. An experiment was carried out to see what factors are associated with the running times of individuals during a new portion of a Tough Mudder obstacle course (in minutes). The finishing time of the various test runners was indicated above in minutes**.**

Min.	Q_1	Median	Mean	Standard Deviation	Q_3	Max
0	5	8.5	9.47	1.2	16	25

    a. Create a boxplot for the data above using the 5 number summary
    b. If the time of a random runner has a z-score of 2, what was the runner's time in minutes?
    c. Are there any outliers in these data? Justify using calculations.

19. The following shows the times (in seconds) required to complete a world cup downhill skiing training run for a random sample of nine skiers. The mean is 50.944 and the standard deviation is 9.837 seconds. The data is shown below (in seconds).

43 43 48 48.5 49 50 50 51 76

    a. What is the median of the data?
    b. What is the value of Q1?
    c. What is the value of Q3?

d. What is the value of the interquartile range?
e. What is the z-score for the skier who took 43 seconds to complete the run?
f. Are there any outliers within the data?

20. During the actual men's world cup downhill ski final in Banff this past year (2019), 37 males competed. The winner was the athlete who got down the hill the fastest. The results of 37 skiers who raced were analyzed. The mean time of all skiers was 108.24 seconds and the standard deviation was 0.626 seconds. The 5 number summary of their results is shown below (time in seconds)

Min	Q1	Q2	Q3	Max
106.81	108.025	108.44	109.2	114.31

a. What is the value of the IQR?
b. Are there any outliers in the data?
c. What is the Z-score of the winner, who skied a time of 106.81 seconds?
d. Is the data skewed?
e. What is the probability that a skier will ski 107 seconds or less?
f. What is the probability that a skier will ski between 107 and 108.5 seconds
g. What is the probability that a skier will ski longer than 110 seconds?
h. What time would a skier need to be in the top 10% of all times (with top being a fast time)?

21. The average number of times that Americans dine out in a week fell from 4.0 in 2008 to 3.8 in 2012. The following is a sample from 20 families that show the number of times they dined out during the week.

6, 1, 5, 3, 7, 3, 0, 3, 1, 3, 4, 1, 2, 4, 1, 0, 5, 6, 3, 1

a) Compute the mean and median
b) Compute the first the third quartiles
c) Compute the range and interquartile range
d) Compute the variance and standard deviation
e) What is the z-score for a family that dined out 5 times per week
f) Are there any outliers in the data? Test using the IQR test
g) Draw a boxplot for the data using the 5 number summary
h) Draw a histogram for the data

22. The following data show the number of hours that workers attained during a recent night for a sample of 20 workers.

6, 5, 10, 5, 6, 9, 9, 5, 9, 5, 8, 7, 8, 6, 9, 8, 9, 6, 10, 8

a) Compute the mean and median
b) Compute the first the third quartiles
c) Compute the range and interquartile range
d) Compute the variance and standard deviation
e) What is the z-score for the worker that slept for 9 hours per night
f) Are there any outliers in the data? Test using the IQR test
g) Draw a boxplot for the data using the 5 number summary
h) Draw a histogram for the data
i) What is the 5 number summary for this data?

23. Annual sales in millions of dollars for 21 pharmaceutical companies are as follows:

8408, 1374, 1872, 8879, 2459, 11413, 608, 14138, 6452, 1850, 2818, 1356, 10498, 7478, 4019, 4341, 739, 2127, 3653, 5794, 8305

a) Compute the mean and median
b) Compute the first the third quartiles
c) Compute the range and interquartile range
d) Compute the variance and standard deviation
e) What is the z-score for the company that made 1872 (millions) of dollars.
f) Are there any outliers in the data? Test using the IQR test
g) Draw a boxplot for the data using the 5 number summary
h) Draw a histogram for the data
i) What is the 5 number summary for this data?

24. The following shows the response time, in minutes, for 10 emergency calls in Burnaby:

11.8, 10.3, 10.7, 11.5, 8.3, 10.5, 10.9 10.7, 11.2

a) Compute the mean and median
b) Compute the first the third quartiles
c) Compute the range and interquartile range
d) Compute the variance and standard deviation
e) What is the z-score for the emergency call that took 11.5 minutes.
f) Are there any outliers in the data? Test using the IQR test
g) Draw a boxplot for the data using the 5 number summary
h) Draw a histogram for the data
i) What is the 5 number summary for this data?

25. The median starting salary (monthly) for 12 graduates from Langara College are noted below.

3710, 3755, 3850, 3880, 3880, 3890, 3920, 3940, 3950, 4050, 4130, 4325

The 5 number summary for the data is the following:

Minimum = 3710; Q1 = 3857.5, Q2 = 3905, Q3 = 4025, Max = 4325. The mean is 3940 and the sample standard deviation is 165.65

a) Compute the range and interquartile range
b) What is the z-score for a salary that was 4130 (average monthly salary).
c) Are there any outliers in the data? Test using the IQR test
d) Draw a boxplot for the data using the 5 number summary

26. The five number summary for samples of commuters in two cities was constructed (in minutes). They are the following:

	Min	**Q1**	**Median**	**Q3**	**Max**
Chicago	5	10	20	30	60
New York	5	15	22.5	42.5	85

a) Draw a boxplot that shows both of the two summaries on the same plot. Comment on the plots.
b) Compute the IQR for each of the two samples of data?

c) Are there any outliers that you can see within the data? What is the formula used to calculate?

27. The 2013-2014 roster of the Seattle Seahawks, winners of the 2014 Superbowl, included 10 defensive linemen and 9 offensive linemen. The weights of the defensive linemen are noted below:

254, 311, 297, 323, 260, 242, 300, 252, 303, 274

On the other hand, the weights of the offensive linemen are noted below:

310, 315, 305, 318, 298, 301, 321, 332, 320

For each of the two samples:

a) Compute the mean and median
b) Compute the first the third quartiles
c) Compute the range and interquartile range
d) Compute the variance and standard deviation
e) What is the z-score for each of the samples for a lineman that weighs 300 pounds.
f) Are there any outliers in the data? Test using the IQR test
g) Draw a boxplot for the data showing both the defensive and offensive linemen.
h) What is the 5 number summary for this data?

28. Many businesses operate call centers to serve customers who want to place an order or make an. Customers want their requests handled thoroughly. Businesses want to treat customers well, but they also want to avoid wasted time on the phone. They therefore monitor the length of calls and encourage their representatives to keep calls short. We have data on the length of all 31,492 calls made to the customer service center of a small bank in a month. To analyze this data question I have taken a sample of 80 calls from the call center. **The five-number summary for these data is 1.0, 54.4, 103.5, 200, 2631. The data is in seconds. The mean is 197.**

a) Create a box-plot for this data using the 5 number summary?
b) Would you describe the data as skewed? Why or why not?
c) Determine whether there are any outliers within the data? Note that the outlier formula is Q1 / Q3 +/- 1.5*IQR

Normal Distribution (For Midterm 1)

29. Suppose that resting pulse rates for health adults are found to follow a normal distribution with a mean of 69 beats per minute and a standard deviation of 9.5 beats per minute.
    a. What is the probability that a random person will have a resting pulse rate of 75 or more?
    b. What is the probability that a random person will have a resting pulse rate of 62 beats per minute or less?
    c. What is the probability that a random person will have a resting pulse rate of between 63 and 71 beats per minute?

30. The length of human pregnancies from conception to birth varies according to the normal distribution with a mean of 266 days and a standard deviation of 16 days.
    a. Almost all pregnancies (99.7%) of pregnancies fall within what length of days?
    b. How long are the longest 2.5% of all pregnancies?
    c. How short at the shortest 16% of pregnancies?
    d. What is the probability that a pregnancy will last 272 days or longer?
    e. What is the probability that a pregnancy will last 250 days or shorter?
    f. What is the probability that a pregnancy will last between 250 and 290 days?

31. Bigger animals tend to carry their young longer before birth. The length of horse pregnancies from conception to birth varies according to a roughly normal distribution with a mean of 336 days an a standard deviation of 3 days.
    a. Between what values do the lengths of the middle 95% of all horse pregnancies fall?
    b. What percentage of horse pregnancies are less than 333 days?

32. The distribution of heights of mean is approximately normal with a mean of 69.2 inches and a standard deviation of 2.5 inches.
    a. What percentage of men are shorter than 61.7 inches?
    b. What percentage of men are larger than 66.2 inches?
    c. What percentage of men are larger than 72.3 inches?
    d. Between what heights do the middle 95% of men fall within?

33. The distribution of hours of sleep per school night, among high school seniors, is found to be normally distributed with a mean of 6.6 hours and a standard deviation of 1.3 hours.
    a. What percentage of high school seniors sleep more than 7.9 hours?
    b. What percentage of high school seniors sleep less than 4 hours?
    c. What is the probability that a high school senior sleeps between 6 and 9 hours of sleep?
    d. What range contains the middle 68% of hours slept per weeknight by high school seniors?

34. The average wind speed in San Fransisco in July was measured in 2019 to be 13.6 miles per hour. The distribution was assumed to be normally distributed with a mean of 13.6 mph and a standard deviation of 4 mph.
    a. What is the probability that, for a random day within the month, the wind speed was 22 mph or more?
    b. What is the probability that, for a random day within the month, the wind speed was less than 6 mph?
    c. What is the probability that, for a random day within the month, the wind speed will be less than 7.2 mph? Greater than 20 mph?
    d. What is the probability that, for a random day within the month, the wind speed was between 8 and 18 mph?
    e. What speed does the wind have to be greater than to be in the top 15% of wind speed for a particular month?
    f. Suppose that, on a particular day in July, there is no wind. Should this be considered unusual?

35. A study of Pennsylvania hospitals showed that the mean patient length of stay was 4.87 days, with a standard deviation of 0.97 days. Assume that the distribution of patient length stays is normally distributed.
    a. What is the probability that a patient will stay for 5 days of longer?
    b. What is the probability that a patient will stay at the hospital for 3 days or shorter?
    c. What is the probability that a patient will stay between 2 and 5 days?
    d. What is the probability that a patient will stay for 6 days or more?

36. Six week old babies consume a mean of 15 ounces of milk per day with a standard deviation of 2 ounces. Assume that the distribution is normally distributed.
    a. What is the probability that a random baby will consume less than 15 ounces per day?
    b. What is the probability that a random baby will consume more than 17 ounces per day?
    c. What is the probability that a random baby will consume between 17 and 19 ounces per day of milk?

37. Statista.com reported in 2014 that the mean number of Facebook friends for 18 to 24 year old is 649 with a standard deviation of 150 friends. Find the probability that a randomly selected Facebook user has the following number of friends.
    a. Fewer than 424 friends?

b. More than 949 friends?
c. Between 424 and 949 friends?
d. Greater than 800 friends?
e. Less than 100 friends?
f. More than 1174 friends?

38. The World Health Organization states that tobacco is the second leading cause of death in the world. Every year, a mean of 5 million people die from tobacco related causes with a standard deviation of 2 million.
    a. Find the probability that more than 4 million people will die of tobacco related causes in a particular year?
    b. What is the probability that less than 1 million people will die of tobacco related causes in a particular year?
    c. What is the probability that between 2 and 7 million people will die of tobacco related causes in a particular year?

39. The College Board recently released a report that the population mean Math SAT score in 2019 was 514, with a population standard deviation of 118. The data was considered to be normally distributed.
    a. Suppose that the college board wants to determine at risk students, especially those who score equal to or less than 321. Determine the probability of a student scoring 321 or less. Draw the graph, shade the area, and calculate the probability.
    b. What is the probability of a student scoring between 215 and 595 on the Math SAT section? Draw the graph, shade the area, and calculate the probability.
    c. We randomly select a student. What is the score that students who score in the top 17% of all students meet? Draw a diagram that depicts your answer and solve.

40. The GMAT is a standardized exam used by many universities as part of the assessment for admission to graduate study in business. The scores are normally distributed, with a mean of 547 and a standard deviation of 100. For every question draw the normal curve and shade the appropriate area:
    a) What is the percentage of scores that are 647 or higher?
    b) What percentage of scores are 747 or higher?
    c) What percentage of scores are between 447 and 544?
    d) What percentage of GMAT scores are between 347 and 647?
    e) What score is the score that 75% of the total writers are equal or less than?

41. The amount of rainfall (in mm/day) at a certain location is well modeled by a normal distribution with mean 17 and Standard Deviation 4.
    a) One day, the amount of rainfall was recorded and its z-score was calculated to be -2.1. How much rain fell on this day? Answer this by interpreting the number -2.1
    b) What is the probability that it will rain between 13 and 22 mm / day in that location?
    c) What is the probability that it will rain less than 12 mm / day in that particular location?

42. A recent study showed that the amount of time student athletes spent in the gym doing cardiovascular work follows a normal distribution with an average of 43 minutes and a standard deviation of 9.2 minutes.
    a. When a student athlete is selected, what is the probability that he (or she) will spend more than an hour (i.e. 60 minutes) in the gym doing cardiovascular work?
    b. When a student athlete is randomly selected, what is the probability that he (or she) will spend between 40 minutes and 50 minutes in the gym doing cardiovascular work?
    c. What value do 22% of all student athletes exceed when using the gym? (The top 22%)

d. Some student athletes do not focus on cardiovascular endurance daily. What time does a student athlete have to spend less than in order to be in the bottom 30% of all athletes. Note that bottom 30% means a slower time.

43. The GRE (Graduate Record Examination) test is an admissions test for graduate schools used in universities across Canada (and the U.S.). The GRE scores are known to follow the normal distribution with mean 500 and standard deviation 100.
    a) What percentage of candidates score above 685?
    b) In a university, students with a GRE score in the top 20% are guaranteed admission. What score must a student get to be guaranteed admission?

44. The energy information administration reported that the mean retail price per gallon of regular grade gasoline was $3.43. Suppose that the standard deviation was $0.10 and that the retail distribution has a normal distribution.
    a) What percentage of regular grade gasoline sold between $3.33 and $3.53 per gallon
    b) What percentage of regular grade gasoline sold between $3.33 and $3.63 per gallon
    c) What percentage of regular grade gasoline sold for more than $3.63 per gallon

45. A company conducted a survey of 13,000 brides and grooms married in the United States, and found that the average cost of a wedding is $29,858. Assume that the cost of a wedding is normally distributed with a mean of $29,858 and a standard deviation of $5,600.
    a) What is the probability that a wedding costs less than $20,000?
    b) What is the probability that a wedding costs between $20,000 and $39,000?
    c) For a wedding to be among the 5% most expensive weddings, how much would it have to cost at minimum?

46. The port of Vancouver is one of the busiest ports in the world. Suppose that the port handles an average of 4.5 million tonners of cargo per week, is normally distributed, and has a standard deviation of 0.82 tonnes.
    a) What is the probability that the port will handle more than 5 million tonnes of cargo per week?
    b) What is the probability that the port handles 3 or more million tonnes of cargo per week?
    c) There is a strike that is held at the port. What is the probability that the port will handle less than 1 million tonnes of cargo per week?
    d) What is the probability that the port will handle between 3 and 5 million tonnes per week.

47. The NCAA estimates that the yearly value of a full athletic scholarship at in-state public universities was $19,000. Assume that the athletic scholarship is normally distributed with a standard deviation of $2,100.
    a) For the 10% of athletic scholarships of least value, how much are they worth?
    b) What percentage of athletic scholarships are valued at $22,000 or more?
    c) For the 3% of athletic scholarships that are most valuable, how much are they worth?

48. Automobile costs continue to rise in Canada, with the average cost of a repair being $367 per repair. Assume that the costs are normally distributed with a standard deviation of $88. Answer the following cost of automobile repairs. Be sure to draw the histogram for each question.
    a) What is the probability that the cost will be more than $540
    b) What is the probability that the cost will be less than $250
    c) What is the probability that the cost will be between $250 and $450
    d) If the cost for your car repair is in the lower 5% of automobile repairs, what is your cost?

49. Many families in Canada use backyard structures for home offices, art studios, and hobby areas (assuming the family owns a single detached home). Suppose that the mean price for a customized, wooden structure is $3100 and the standard deviation is $1200.
    a) What is the z-score for a backyard structure costing $2300?
    b) What is the z-score for a backyard structure costing $4900?
    c) What is the probability that the backyard structure will cost $3500 or more?

Chi-Squared Test & Related Probability (For Midterm 1)

50. Many retail chains offer credit cards to repeat customers of their stores as a means to keep them coming back. This way the store gets to build a relationship with its customers that includes an opportunity to track what they purchase with the card. A customer can sign up for a credit card at the time of a purchase in the store, or online, or by mailing in an application. A credit-counselling bureau wishes to investigate whether the method used to recruit a customer for a store credit card is related to subsequent status of the customer's account. The data gathered for 795 eighteen-months-old store credit card accounts are summarized for the two variables of interest in the following table.

	Recruiting Method			
Account Status	**In Store**	**Mail In**	**Online**	**Total**
Closed	55	11	34	**100**
Current	286	77	129	**492**
Late Payment	88	52	63	**203**
Total	**429**	**140**	**226**	**795**

    a) What graph would you use to represent the data in the above two-way table? Do NOT draw the graph.
    b) Clearly state the null and alternative hypotheses in this problem.
    c) Obtain the expected count of the entry in that table for the category "Online and Late Payment" and briefly explains what your answer represents in the context of a chi square investigation.
    d) The retail chain calculated the Chi-Square statistic by hand, and found a Chi-Square statistic of -2.38. He knew that this answer was not correct immediately. Explain (in one sentence) what is wrong with this Chi-square statistic.

51. While online financial transactions are becoming common place, security and privacy issues have also become a concern. A consumer survey investigates whether such concerns are related to age. One of the survey questions asks whether respondents agree or disagree with the statement "I don't feel comfortable giving my credit card number and other personal information online." The possible answers are "Strongly Agree," "Agree," "Disagree," or "Strongly Disagree." The contingency table shown summarizes the data.

Table of Observed Frequencies

	Strongly Agree	*Agree*	*Disagree*	*Strongly Disagree*	*Row Total*
Ages 18-29	*65*	*78*	*82*	*20*	*245*
Ages 30-49	*72*	*69*	*41*	*36*	*218*
Ages 50-65	*92*	*50*	*28*	*21*	*191*
Ages 65+	*95*	*68*	*20*	*16*	*199*
Column Total	*324*	*265*	*171*	*93*	*853*

a. We want to use the Chi-square procedure to examine whether OPINION is associated with AGE. In the context of the problem, state the Alternative Hypothesis. [1 mark]
b. Calculate the Expected Value (Count) for the cell "Strongly Agree" and "Ages 65+", and provide an interpretation in the context of the problem. [3 mark]
c. The sample association has been found to be statistically significant. If this is so, the Chi-square statistic $(\chi 2)$ must be equal to (check one) □ 0, □ 15.51, □ 66.58. Explain your reasoning. [3 marks]

52. The Pew internet and American life project examined whether single men and women differed with respect to their current relationship. A study was conducted to explore whether the reported type of relationship depends on gender. The table of observed frequencies was noted below.

Type Of Relationship	**Single Men**	**Single Women**
In committed relationship	115	138
Not in committed relationship and not looking for a partner	162	391
Not in a committed relationship but looking for a partner	89	54
Don't know / refused	19	18

a) Calculate the expected value for the number of single men who are in a committed relationship?
b) Calculate the expected value for the number of single women who are not in a committed relationship and not looking for a partner.
c) Write out the null and alternative hypothesis for this particular problem?
d) Suppose the chi-squared value for this problem is 61.12. Based on the degrees of freedom for this table, would you accept or reject the null hypothesis?

53. A study was recently conducted in the US on children 12 – 15 years old. Three categories of children were measured, with an information being compared to how much time they spend in front of the computer. The data for the study was gathered.

	Normal or underweight	**Overweight**	**Obese**
Using computer more than two hours per day	114	28	52
Using the computer two hours or less per	355	96	121

day			

a) Calculate the expected value for the number of children who are normal or underweight and who use the computer more than two hours per day?
b) Calculate the expected value for the number of children who are obese and use the computer two hours or less per day?
c) Write out the null and alternative hypothesis for this particular problem?
d) Calculate the chi-squared value for this problem. What does this mean in relation to the null hypothesis?

54. The center for disease control tracked the number of deaths due to weather related causes. Is there a difference in the pattern of deaths in young and older individuals? The following table summarizes the data.

Age Group	Heat-Related	Cold-Related	Floods / Storms / Lightning
15-24	106	286	97
75-84	198	372	174

a) Calculate the expected value for each of the cells in this problem?
b) Write out the null and alternative hypothesis for this particular problem relating to a test of independence?
c) Calculate the chi-squared value for this problem. What does this mean in relation to the null hypothesis?

55. A study was conducted on college students to better understand their living arrangements. For this data, a sample of 200 young people aged 1 – 24 where asked to indicate their gender and their living arrangements. The following data was collected:

	Living With Parents	Living With Partner	All Other Arrangements
Female	51	22	28
Male	58	14	27

a) Calculate the expected count for the top right cell (i.e., 'Female' and 'Living With Parents'), and explain what the number means in words.
b) Calculate the expected value of the cell corresponding to the number of males who are living with parents.
c) Suppose a chi-square test was done and the value of the test statistic, χ^2, was 2.226. Can the we conclude that an association exists between the gender of college students and their living arrangement? Show all steps of the hypothesis test and use an appropriate decision point to support your conclusion. Clearly state the null and alternative hypothesis. (5 marks)
d) If a person is randomly selected from the study, what is the probability that the person was 'female' or is 'living with a partner'?
e) Suppose a person is randomly selected from the study. What is the probability that a person is living in 'all other arrangements' and a 'male'?
f) Suppose a customer is randomly selected from the study. What is the probability that, given that they were 'female', they were very 'living with parents'?

56. The ski patrol group for Cypress mountain wanted to better understand the relationship between hill level and fall frequency on hills at Cypress Mountain. They took a sample of 1789 individuals over the weekend to look at injury frequency, with the data being shown below.

	Falls On Run	
Hill Difficultly	**Yes**	**No**
Green (Easiest)	431	843
Blue (Intermediate)	51	242

Black (Advanced)	21	103
Double Black (Expert)	7	91

a) State the null and alternative hypothesis for this problem?
b) What graph and statistic will you use to solve it?
c) Calculate the expected value for the number of people who will fall on a green run?
d) Calculate the expected value for the number of people who will fall on a double black run?
e) Calculate the chi-squared value for the number of people who will not fall on a black run (Black & No)
f) The Chi-squared value for this example was 65.52. What does this mean for your null hypothesis?
g) What is the probability that someone falls on a double black run?
h) What is the probability that someone will fall from the sample?
i) What is the probability that someone will attempt a green run from the sample?
j) What is the probability that someone will fall and be on a green run?
k) What is the probability that someone will fall or be on a double black run?
l) Given that someone is skiing or snowboarding on a black run, what is the probability that they will fall?
m) Given that someone is on a green run, what is the probability that they will fall?

57. A market research company was interested in comparing three communities A, B and C to determine whether there are any differences in their use of water filters in homes. Each of 100 homeowners sampled from each community was asked whether or not they have installed water-filtering system in their homes. Their responses (Yes or No) are summarized in the following contingency table.

	Use filter?	
Community	**Yes**	**No**
A	62	38
B	58	42
C	73	27

a) Clearly state the null and alternative hypotheses in this problem.
b) Calculate the expected count for the bottom right cell (i.e., Community 'C' and 'No'), and explain what the number means.
c) The Chi-square statistic has been calculated to be 13.22. What can we conclude from this finding?

58. A sports injury expert sociologist gathered data on 407 individuals skiing on a local North Shore mountain (Cypress Mountain) to look at the relationship between the difficulty of a run and the number of confirmed injuries on the run during a one day period. For each run type, she determined the number of injuries and categorized them into either injured or not. These injuries were confirmed by the ski patrol after the ski patrol stopped to intervene. Some people who they helped (the no group), were not ultimately injured. The run types were either green (easy), blue (intermediate), black (advanced) or double black (expert). The observed data is shown in the following summary table:

Run Type	**Injuries Confirmed**	
	Yes	**No**
Green (Easy)	31	71
Blue (Intermediate)	20	82

Black (Advanced)	5	94
Double Black (Expert)	1	103

a. What is the alternative hypothesis for this question?
b. What is the expected count for those who are injured (Injury Observed = Yes) and those on a green run?
c. What is the expected count for those who are injured (Injury Observed = Yes) and those on a black run?
d. What is the chi-squared value for the cell showing those who are injured (Injury Observed = Yes) on a blue run?
e. How many degrees of freedom does the table have?
f. If the chi-squared value was computed to be 46.68 for this table, would you accept or reject the null hypothesis for the chi-squared test of independence?
g. What is the probability that a skier skied was on a green run?
h. What is the probability that a skier had a confirmed injury?
i. What is the probability that a skier had a confirmed injury and skied on a black (advanced) run?
j. What is the probability that a skier had a confirmed injury or skied on a green run?
k. Given that a skier skied on a double black run, what is the probability that a skier was injured?

59. The following contingency table shows 200 subjects classified as obese or not obese, and as having a mother who did or did not smoke during pregnancy.

	Obese	
Smoking	**Yes**	**No**
Mother Smoked	20	35
Mother did NOT smoke	26	119

a) If we are testing the independence between smoking and obesity, clearly state the null and alternative hypotheses in this problem.
b) Calculate the expected count for the bottom right cell (i.e., 'Mother did Not smoke' and 'No'), and explain what this number means.
c) The Chi-square statistic has been calculated to be 60.05. What inference would you make based on this value of Chi-square? Justify your answer.

60. A department store tracks the payment methods that are used in the store and the various levels of the transactions. We are interested in exploring the relationship between spending and whether cash was used or not. The data are summarized in the following table:

	Using Cash	
Spending	**Cash**	**No Cash**
Under $95	48	18
$95 and over	10	45

a) If we are testing the independence between spending and using cash, clearly state the null and alternative hypotheses in this problem.
b) Calculate the expected count for every cell in the table.
c) Calculate the chi-squared value for the problem in question.

61. A sociologist gathered data on 25 weddings in her local church over a six month period. For each wedding, she determined the Cost (Low= $25,000 or less, High=more than $25,000) and the Payer, as shown in the following summary table. Observed counts for the data are shown below.

PAYER	COST	
	High	**Low**
Bride & Groom	6	3
Bride's Parents	4	10
Groom's Parents	0	2

a. What is the null hypothesis for this question?
b. What is the expected count for those whose payer is the Bride & Groom and the cost is high?
c. What is the expected count for those whose payer is the Groom's parents and the cost is low?
d. What is the chi-squared value for those where the bride's parents pays and the cost is low?
e. How many degrees of freedom does the table have?
f. If the chi-squared value was computed to be 23.516 would you accept or reject the null hypothesis for the chi-squared test of independence?
g. What is the probability that a bride & groom paid for the wedding?
h. What is the probability that a bride's parents paid for the wedding and the cost was high?
i. What is the probability that a groom's parents paid for the wedding or the cost was low?
j. Given that the bride's parents paid for the wedding, what is the probability that the cost was low?

62. A study was conducted to answer the following question: "In the past 12 months, when travelling for business, what type of airline ticket did you purchase?"

	Type Of Flight	
Type Of Ticket	**Domestic**	**International**
First class	29	22
Second class	95	121
Third class	518	135

a) Calculate the expected value for the table using the chi-squared approach.
b) Calculate the value of the chi-squared statistic for this problem.
c) Is the type of ticket independent of the type of flight?
d) State the null hypothesis and the alternative hypothesis that you used for your test?

63. A company surveyed a cross section of North American companies to understand how human resources executives planned to change their workforce over the next 12 months. They surveyed private and public companies and grouped the changes into three categories. The sample data for all companies is noted below:

	Company	
Employment Plan	**Private**	**Public**
Add Employees	37	32
No Change	19	34
Lay Off Employees	16	42

a) Discuss any differences in the employment plans for private and public companies.
b) Conduct a test of independence to determine if the employment plan for the next 12 months is independent of the type of company. At a 0.05 level of significance, what is your conclusion?

64. Health insurance benefits often vary by the size of the company. The sample data shows the number of companies providing health insurance for small, medium, and large companies. Small companies have less than 100 employees, medium sized employees have 100 to 999 employees, and large companies have 1000 or more employees. The following is the data from the test:

	Size Of The Company		
Health Insurance	**Small**	**Medium**	**Large**
Yes	36	65	88
No	14	10	12

a) Conduct a significance test to determine whether health insurance coverage is independent of the size of the company. What is the p-value? What is the value of the chi-squared statistic for this test? Using a 0.05 level of significance, what is your conclusion?

65. The following data shows promotion data from the Vancouver police department.

	Gender	
Promotion Status	**Male**	**Female**
Promoted	288	36
Not Promoted	672	204

a) Conduct a significance test to determine whether being promoted is independent from gender within the Vancouver police department? What is the p-value? What is the value of the chi-squared statistic for this test? Using a 0.05 level of significance, what is your conclusion?

66. Below are the results of a survey of individuals who were asked if they used social media and other websites to voice their opinions about world events.

	Used social medial and other websites to voice their opinion about world events?	
Gender	**Yes**	**No**
Male	395	291
Female	323	355

a) Conduct a significance test to determine whether being gender is independent from whether someone uses social media and other websites to voice their opinion about world events? What is the p-value? What is the value of the chi-squared statistic for this test? Using a 0.05 level of significance, what is your conclusion?

67. A study was conducted to investigate if the level of education (elementary, secondary, or post secondary) is

related to awareness of safe sex practices (not aware, somewhat aware, or very aware).

a. What statistical technique is the most appropriate for the study?
b. If the chi-squared statistic is equal to 15.2, what conclusion should be made?
c. What condition(s) must be satisfied for the above analysis to be valid?

Probability (Midterm 1)

Probability is a numerical measure of the likelihood that an event will occur. Probability values are always assigned on a scale of 0 to 1. The probability of any outcome of a random phenomenon is the proportion of times the outcome would occur in a very long series of repetitions.

We call a phenomenon **random** if individual outcomes are uncertain but there is nonetheless a regular distribution of outcomes in a large number of repetitions.

A **random variable** is a variable whose value is a numerical outcome of a random phenomenon.

The **probability distribution** of a random variable X tells us what values X can take and how to assign probabilities to those values.

There are 4 key probability rules. They are:

- Any probability is a number between 0 and 1 [0 <= P(A) <= 1]
- All possible outcomes together must have probability 1
- If two events have no outcomes in common, the probability that one or the other occurs is the sum of their individual probabilities. [P(A or B) = P(A) + P(B)]. ***Disjoint***.
- The probability that an event does not occur is 1 minus the probability that the event does occur. P(A does not occur) = 1 – P(A). ***Complement***.

Two events are **independent** if knowing that one occurs does not change the probability that the other occurs. If A and B are independent, P(A and B) = P(A)P(B)

68. The trustee of a company's pension plan has solicited the opinion the company's employees about a proposed investment the plan. A breakdown of the responses is shown below:

	Employee Type			
Response	Technicians	Administrators	Others	Total
Support	31	13	16	60
Oppose	24	16	15	55
Total	55	29	31	115

a. If one employee is selected at random, what is the probability that he/she is not an administrator.
b. If one employee is selected at random, what is the probability that he/she is either an Administrator, or supports the proposal.
c. Are the events “selecting an administrator” and “selecting an employee who supports the proposal” mutually exclusive? Explain your answer.

69. Below are the results of a survey of individuals who were asked if they used social media and other websites to voice their opinions about world events.

	Used social medial and other websites to voice their opinion about world events?	
Gender	**Yes**	**No**
Male	395	291
Female	323	355

a) Create a joint probability table for the data in question?
b) What is the probability that the respondent is a female?
c) What is the probability that the respondent is not a female?
d) What is the conditional probability that, given that a respondent uses social media and other websites to voice their opinion about world events, they are male?
e) What is the probability that a respondent is a female or uses social media and other websites to voice their opinion about world events?

70. A telephone survey works to determine viewer response to a new television show! The following are the responses from the survey.

Rating	**Frequency**
Poor	4
Below Average	8
Average	11
Above Average	14
Excellent	13

a) What is the probability that a randomly selected viewer will rate the new show as average or better?
b) What is the probability that a randomly selected viewer will rate the new show as average or worse?

71. The following data shows promotion data from the Vancouver police department.

	Gender	
Promotion Status	**Male**	**Female**
Promoted	288	36
Not Promoted	672	204

a) What is the probability of getting promoted within the department?
b) Given the a randomly selected officer was promoted, what is the probability that they are female?
c) What is the probability of being promoted and being male?
d) What is the probability of being promoted or being female?
e) Given that a randomly selected officer was female, what is the probability that they will be promoted?

72. The Biology Department at a university plans to recruit a new faculty member. Data collected by a different university on the 410 possible candidates were available. The Biology Department is debating whether to put a requirement of 10 years of teaching experience in the job advertisement. The available data on the candidates are shown below:

	Work Experience	
Gender	**Less than 10 years experience**	**10 or more years experience**
Male	178	112
Female	99	21

a) What percentage of candidates is female or has less than 10 years experience?
b) What is the probability that the selected candidate has less than 10 years' experience and is female?
c) What is the probability that candidate is a male or has more than 10 years of experience?
d) Given that a candidate has 10 or more years experience, what is the probability that they are female.

Confidence Intervals (Midterm 2)

- Confidence interval for the mean of a Normal population (draw an SRS of size n from a Normal population having unknown mean μ and known standard deviation σ: $\bar{x} \pm z\frac{\sigma}{\sqrt{n}}$, where z~N(0,1)
- Sampling distribution of a sample proportion: when the population and the sample size are both large and p is not close to 0 or 1, $\hat{p}$ has approximately Normal distribution with mean p and standard deviation $\sqrt{p(1-p)/n}$
- Large-sample z confidence interval for p: $\hat{p} \pm z\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$, where z~N(0,1)

73. The average cost of the first year of owning and caring for a large dog is very high. The Irish Red and White Setter Association conducted a study to estimate the annual first year cost for owners of this breed. A sample of 50 families was used and the cost was found to be a sample mean of $1,843. Based on past studies, the population standard deviation is assumed known with a standard deviation of $255.
    a. Construct a 68, 95, and 95 confidence interval for the mean cost of first year dog ownership.

74. The manager at a large fitness center keeps track of the amount of time it takes a sample of 65 clients to run a mile. She finds that the mean time to run a mile in this sample is 10.4 minutes, with the standard deviation of 2.1 minutes. Assume that the sample is a random sample that is normally distributed.
    a. Construct a 68, 95, and 99 confidence interval for the parameter of interest in this problem.

75. A study was recently conducted to review the fuel efficiency of motor vehicles. On one study, the main variable of interest was the MPG of vehicles. For this study, 100 vehicles were surveyed, with a sample mean city gas mileage of 20.71 mpg (miles per gallon). The standard deviation was 5.637.
    a. Construct a 68, 95, and 99 confidence interval for the parameter of interest in this problem.

76. A psychology study of 36 boys found that the mean length of time that boys remained engaged with a science exhibit at a museum was 107 seconds with a standard deviation of 117 seconds for the population.
    a. Construct a 72, 90, and 97% confidence interval for the parameter of interest in this problem.

77. In a famous research paper from psychology literature, George Miller found that the average amount of information humans could process in short-term memory was 7 bits (pieces of information), with a standard deviation of 2 bits. In this case n = 30 (the number of individuals currently being sampled).
    a. Construct a 90, 95, and 99 percent confidence interval for the average amount of information that humans can process.

78. A randomized comparative experiment studied the effect of diet on blood pressure. Researchers divided 54 healthy white males at random into two groups. The average systolic blood pressure of the 27 members of the placebo group was reported to have a mean of 114.9 and a standard deviation of 9.3.
    a. Give a 95% confidence interval for the mean blood pressure of the placebo group.

79. A recent survey found that 697 of a sample of 1041 adults said that they wanted stricter laws covering the sale of firearms.
    a. Calculate an interpret a 90% and 95% confidence interval for the parameter of interest?

80. In 2016, a Pew Center research survey on public library usage found that 768 adults out of a sample of 1601 had visited, in person, a library or bookmobile in the last year.
    a. Calculate an interpret a 82% and 70% confidence interval for the parameter of interest?

81. Shoplifting is an escalating problem for retailers. A New York City store chose 500 shoppers at random and monitored them while they were in the store. Forty of them were caught shoplifting.
    a. Calculate the 90% confidence interval for the proportion of all shoppers who shoplift.
    b. Interpret the above confidence interval in words.

82. A Pew Research Center survey of 1895 users found that 1118 agreed that "online dating is a good way to meet people."
    a. Construct a 90, 95, and 99% confidence interval for the parameter of interest in this problem.
    b. Interpret the above confidence interval in words.

83. Millenials refers to the generation of young people aged 18 – 29 in 2010. In the year 2010, a Pew Research Center study found that 183 out of a sample of 830 American millennials were married.
    a. Construct an 86, 90, and 92% confidence interval for the proportion of American millennials who were married in 2010. Interpret the confidence interval in words.

84. An NHANES report gives data for 654 women aged 20 to 29 years. The mean BMI of these 654 women was 26.8. On the basis of the sample, estimate the mean BMI in the population of all 20.6 million women in the age group. Let us assume that the sample is from a normal population with standard deviation of 7.5.

    a) Construct a 95% confidence interval for mean BMI of all women
    b) Construct both a 90% and 99% confidence interval for the mean BMI in this population?
    c) How does increasing the confidence level change the margin of error of a confidence interval when the sample size and population standard deviation remain the same?
    d) Suppose that the survey above had a sample size of just 100 women. What is the 95% confidence interval for this situation? Is it different from the initial case?
    e) Construct a 95% confidence interval for samples with 400 and 1600 women. How are these confidence intervals different?

85. Does the expectation of good weather lead to more generous behavior? Psychologists have studied the size of the tip in a restaurant when a message indicates that the next day's weather would be good was written on the bill. After surveying 20 patrons, they found that the mean tip, in percent of the total bill, was 22.21 percent, with a standard deviation of 2 percent.

    a) Estimate the mean tip percentage for all patrons of the restaurant when they receive a message on their bill that says the next days weather will be good. Use a 95% confidence interval for this score.

86. There are many people who would rather be active outdoors, especially in certain states. In 2019, a study in Minnesota found that 39 out of 101 individuals surveyed would rather be fishing outdoors than be at work indoors. This proportion led the nation within the category of fishing.

a. Construct a 90, 95, and 99% confidence interval for the parameter of interest in this problem. What are your conclusions?

87. A study released by the American Medical Association in 2019 found that 83% of college female students agreed that heavier drinking occurs on spring break trips than is typically found on campus. Assume that the sample size for this study was 103 individuals.
    a. Construct an 80, 90, and 95% confidence interval for the parameter of interest in this problem. What are your conclusions or what is your interpretation for these problems?

88. American Demographics magazine reported that 40% of a sample of NASCAR racing attendees said that they owned a pickup truck. Suppose that the sample size for this study was 904. Construct a 90% confidence interval for the proportion of NASCAR drivers within the population who own a pickup truck.

89. According to a Pew Research Center Study, out of 99 respondents to a study on cell phone ownership and closeness, 21 of those surveyed noted that they have felt closer in relationships to their spouse or partner because of exchanges that they have had online or via text message.
    a. Construct an 80, 90, and 95% confidence interval for the parameter of interest in this problem. What are your conclusions or what is your interpretation for these problems?

90. A class survey in a large class for first year college students asked "About how many hours do you study during a typical week?" The mean response of the 463 students was 15.3 hours. Suppose that we know that the study time follows a normal distribution with standard deviation of 8.5 hours in the population of all first year students at this university.

    a) Construct a 90, 95, and 99% confidence interval for the mean study time of all first year students.
    b) Assume that there were actually 464 students who responded, with one student claiming to study 10,000 hours a week. This value was clearly a joke and so it was omitted. Leaving the 10,000 hours in the data, we would get a mean of 36.8 hours for the 464 students and a standard deviation of 8.5. Construct new 90, 95, and 99 percent confidence intervals using this new information.

91. A simple random sample of 50 items from a population with a standard deviation of 6 resulted in a sample mean of 32. Provide a 90, 95, and 99% confidence interval for the population mean.

92. Studies have shown that massage therapy has a variety of health benefits and it is not too expensive. A sample of 10 typical one-hour massage therapy sessions showed an average charge of $59 and a population standard deviation for a 1 hour session of about $5.50.

    a) Create a 90, 95, and 99% confidence interval for the population mean of all clients who get a massage

93. Costs are rising for all kinds of medical care. The mean monthly rent at assisted-living facilities was reported to have increased 17% over the last 5 years to $3486. Assume that this cost is based on a sample of 120 facilities, and that, from past studies, it can be assumed that the population is normally distribution with a standard deviation of $650.

    a) Develop a 90, 95, and 99% confidence interval estimate for the population mean monthly rent?
    b) What happens to the width of the confidence interval as the confidence level is increased? Does this seem reasonable?

94. A national survey of 900 women golfers was conducted to learn how golfers view their treatment at golf courses in the United States. The survey found that 396 of the women golfers were satisfied with the availability of tee times.
    a) Develop a 90, 95, and 99% point estimate for the proportion of women golfers who are satisfied with their tee times.

95. According to statistics reported on the CBC, a surprising number of motor vehicles are not covered by insurance. Sample results, consistent with the CBC report, show that 46 out of 200 vehicles were not covered by insurance.
    a) Develop a 95% confidence interval for the population proportion of motor vehicles not covered by insurance.

96. The consumer reports national research center conducted a phone survey of 2000 adults to lean about the major economic concerns for the future. The survey results showed the 1760 of the respondents think that the future health of social security is a major economic concern.
    a) Develop a 90, 95, and 99% confidence interval for the population proportion of adults who things that the future health of social security is a major economic concern.

97. According to Franchise Business Review, over 50% of all food franchises earn a profit of less than $50,000 per year. In a sample of 142 casual dining restaurants, 81 earned a profit of less than $50,000 last year.
    a) Develop a 95% confidence interval for the proportion of casual dining restaurants that earned a profit of less than $50,000 last year.

98. A survey conducted by the BCAA found that a family of four spends an average of $215.60 per day while on vacation. Suppose that a sample of 64 families of four vacationing at Niagara falls resulted in a sample mean of $252.64 per day and a sample standard deviation of $74.50.
    a) Develop a 95% confidence interval estimate for the mean amount spent per day by a family of four visiting Niagara Falls.

99. A sample of 54 discount brokers showed that the mean price charged for a trade of 100 shares at $50 per share was $33.77. The survey is conducted annually. With the historical data available, the population standard deviation was estimated to be $15.
    a) Develop a 95% confidence interval for the mean price charged by discount brokers for a trade of 100 shares at $50 per share.

100. The 92 million Americans of age 50 and over control 50 percent of all discretionary income. AARP estimates that the average annual expenditure on restaurants and carryout food was $1873 for individuals in this same age group. Suppose that this estimate is based on a sample of 80 persons, and that the sample standard deviation is $550.
    a) Develop a 95% confidence interval for the population mean amount spent on restaurants and carry out food?
    b) If the amount spent on restaurants and carry out food is skewed to the right, would you expect the median amount spent to be greater or less than $1873?

101. A survey in Canada of 369 working parents found that 200 said they spend too little time with their children because of work commitments.

a) What is the point estimate of the proportion of the population of working parents who feel they spend too little time with their children because of work commitments?
b) Develop a 90, 95, and 99% confidence interval estimate for the population proportion of working parents who feel that they spend too little time with their children because of work commitments?

102. A branch library wishes to estimate the number of books on average it checks out to patrons a day. A study of book circulation at the branch shows that on 60 randomly selected days it opens, an average of 480 books per day, and a standard deviation of 100 books per day are checked out.
   a. Briefly describe the population and the parameter of interest.
   b. Provide a 95% interval estimate of the long run mean number of books per day the library checks out a day.

103. A sample of 1081 hospitalized patients in China, with laboratory-confirmed Covid-19, was obtained. Fever was present in 473 of these patients when they were first admitted to the hospital (Source: New England Journal of Medicine, February 28, 2020).
   a. Calculate a 95% confidence interval for the population proportion (p) of Covid-19 patients that have a fever when they are admitted to the hospital.
   b. Provide an interpretation (in context) of your interval estimate in part (a).

Regression (Midterm 2)

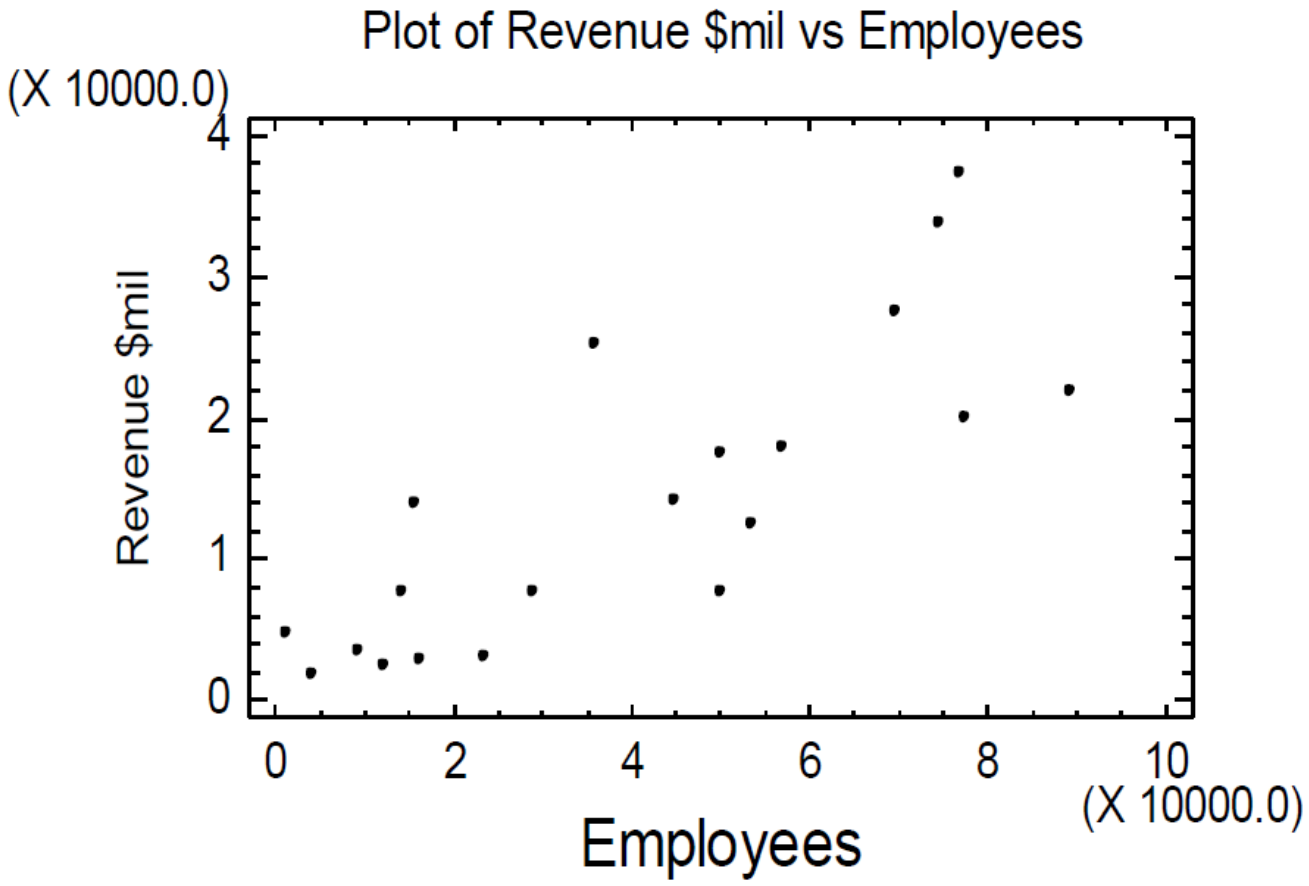
104. What is the relationship between a company's revenue and the size of its work force? In a study involving twenty selected companies, a management research team obtained data for the annual revenue (in $ millions) and the number of people the company has hired during the past three quarters. Carefully study the following analyses of the data and answer the questions.

Simple Regression - Revenue vs. Employees
Dependent variable: Revenue $mil (in $ million)
Independent variable: Employees

Coefficients

Parameter	Least Squares Estimate	Standard Error	T Statistic	P-Value
Intercept	1292.67	2554.2	0.506095	0.6189
Slope	0.316516	0.052578	6.01994	0.0000

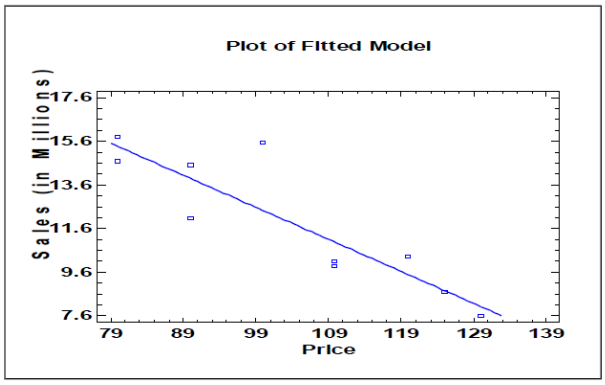
Correlation Coefficient = 0.817398
R-squared = 66.814 percent
R-squared (adjusted for d.f.) = 64.9703 percent
Standard Error of Est. = 6378.72



Answer the following questions, giving all answers in the context of the particular problem.

a. Write down the equation of the estimated regression line.
b. What is the value of the slope of the regression line? Explain what the number means in terms of the variables
c. Is there a significant linear relationship between revenue & the number of employees? Use an appropriate decision point and show all necessary steps to answer this question. Clearly state the null & alternative hypothesis.
d. Use the least squares line to predict the annual revenue of a company with a work force of 30,000. Include a margin of error for your prediction that is valid 95% of the time.

105. Data are collected to analyze the relationship between sales (in millions of units) of a consumer electronic device and its price (in dollars) in 10 retail chains. The output below summarizes the findings.



Simple Regression - Sales (in Millions) vs. Price
Dependent variable: Sales (in Millions)
Independent variable: Price
Linear model: Y = a + b*X

Coefficients

Parameter	*LS Estimate*	*Std. Error*	*T Statistic*	*P-Value*
Intercept	27.14	2.76209	9.82535	0.0000
Slope	-0.15	0.026329	-5.58509	0.0005

Analysis of Variance

Source	*Sum of Squares*	*Df*	*Mean Square*	*F-Ratio*	*P-Value*
Model	64.963	1	64.963	31.19	0.0005
Residual	16.6608	8	2.0826		
Total (Corr.)	81.6238	9			

Correlation Coefficient = -0.892123
R-squared = 79.5883 percent
R-squared (adjusted for d.f.) = 77.0368 percent
Standard Error of Est. = 1.44312

a. Can we conclude that SALES and PRICE are linearly related? Justify your answer by referring to the appropriate decision point and graph. [4 marks]
b. Construct a Prediction Interval for SALES when PRICE is $100. Construct your interval so it will be correct 95% of the time. (Note, the software calls RMSE "Standard Error of Est.") [3 marks]
c. Explain what the Homoscedasticity (constant variance) assumption means in the context of the problem, and explain why this assumption is necessary to validate the Prediction Interval you calculate in part (b) above. [4 marks]

106. The NFL Combine is an important, standardized test that is conducted for over 300 NFL hopefuls every year. During the combine, each individual is required to do a set of specific exercises, with each of their results measured. Below is the regression obtained for the 40 yard dash vs weight at the 2019 NFL combine.

Simple Regression - 40_Yard_Dash vs. Weight

Dependent variable: 40_Yard_Dash
Independent variable: Weight
Linear model: Y = a + b*X
Number of observations: 268

Coefficients

	Least Squares	*Standard*	*T*	
Parameter	*Estimate*	*Error*	*Statistic*	*P-Value*
Intercept	3.44266	0.0492244	69.938	0.0000
Slope	0.00532765	0.000200809	26.531	0.0000

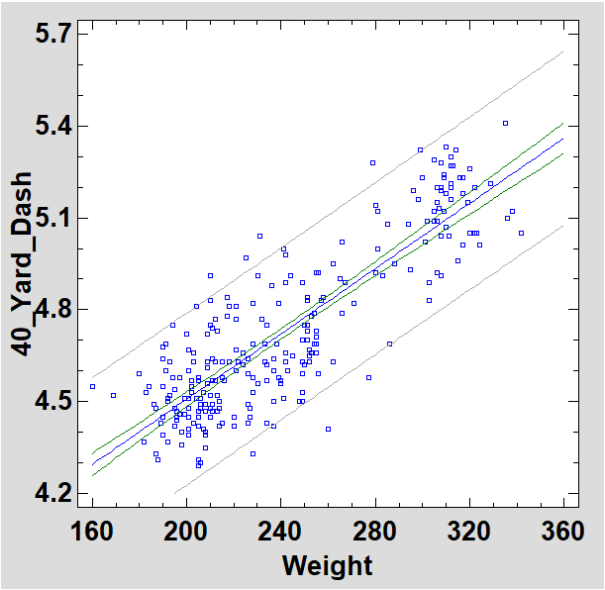
Analysis of Variance

Source	*Sum of Squares*	*Df*	*Mean Square*	*F-Ratio*	*P-Value*
Model	14.2076	1	14.2076	703.89	0.0000
Residual	5.36902	266	0.0201843		
Total (Corr.)	19.5766	267			

Correlation Coefficient = 0.851905
R-squared = 72.5743 percent
R-squared (adjusted for d.f.) = 72.4712 percent
Standard Error of Est. = 0.142071
Mean absolute error = 0.115319
Durbin-Watson statistic = 1.89999 (P=0.2070)
Lag 1 residual autocorrelation = 0.0491508

The scatterplot of the data is also shown below.

a. Determine the explanatory variable, response variable, and list any lurking variables
b. Identify and interpret the correlation coefficient according to the 3 metrics that we discussed
c. Write out the equation of the regression line that fits the model
d. Interpret the slope for the regression line in question.
e. For an individual who weight 273 pounds, what is their predicted 40 yard dash time?
f. Create a 95% confidence interval for the prediction for part e.
g. Write out the null and alternative hypothesis and either reject or accept the null? Complete and write out all steps of the hypothesis test and write a concluding statement.



107. Following this test, at the 2019 NFL combine, evaluators wanted to also test the relationship between the heights of individuals in attendance and their ability to run the 40 yard dash. The following regression output was obtained.

Simple Regression - 40_Yard_Dash_2 vs. Height

Dependent variable: 40_Yard_Dash_2
Independent variable: Height
Linear model: Y = a + b*X
Number of observations: 268

Coefficients

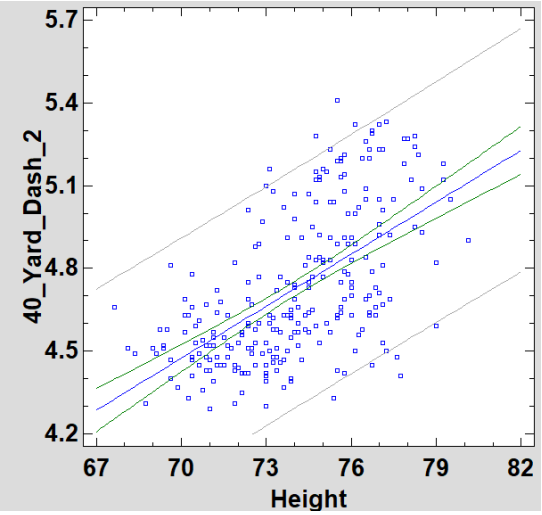
	Least Squares	*Standard*	*T*	
Parameter	*Estimate*	*Error*	*Statistic*	*P-Value*
Intercept	0.0792522	0.392033	0.202157	0.8399
Slope	0.0627947	0.00529225	11.8654	0.0000

Analysis of Variance

Source	*Sum of Squares*	*Df*	*Mean Square*	*F-Ratio*	*P-Value*
Model	6.77541	1	6.77541	140.79	0.0000
Residual	12.8012	266	0.0481248		
Total (Corr.)	19.5766	267			

Correlation Coefficient = 0.5883
R-squared = 34.6097 percent
R-squared (adjusted for d.f.) = 34.3639 percent
Standard Error of Est. = 0.219374
Mean absolute error = 0.177926
Durbin-Watson statistic = 2.1825 (P=0.9322)
Lag 1 residual autocorrelation = -0.0919517

The scatterplot of the data is also shown below.



a. Determine the explanatory variable, response variable, and list any lurking variables
b. Identify and interpret the correlation coefficient according to the 3 metrics that we discussed
c. Write out the equation of the regression line that fits the model
d. Interpret the slope for the regression line in question.
e. For an individual who has a height of 74 inches, what is their predicted 40 yard dash time?
f. Create a 95% confidence interval for the prediction for part e.
g. Write out the null and alternative hypothesis and either reject or accept the null? Complete and write out all steps of the hypothesis test and write a concluding statement.

108. Following this test, during the 2019 NFL combine, the evaluators wanted to test the relationship between the weight of individuals and their ability to vertically leap, measured in inches. The following regression output was obtained.

Simple Regression - Vert_Leap_Inches vs. Weight_2_Lbs

Dependent variable: Vert_Leap_Inches
Independent variable: Weight_2_Lbs
Linear model: Y = a + b*X
Number of observations: 272

Coefficients

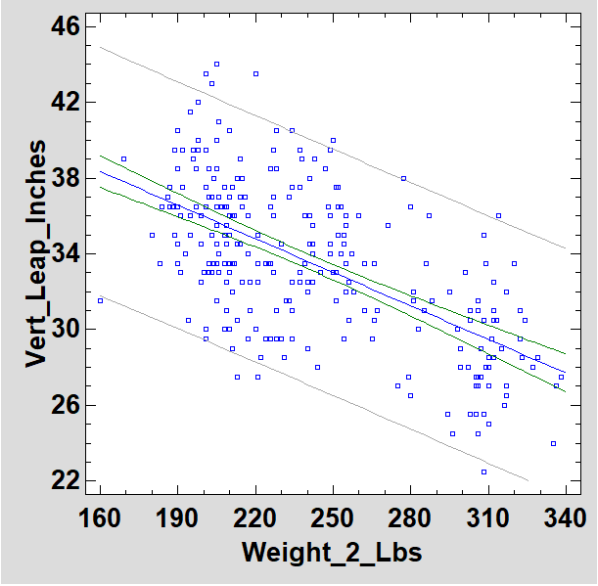
	Least Squares	*Standard*	*T*	
Parameter	*Estimate*	*Error*	*Statistic*	*P-Value*
Intercept	47.8178	1.15361	41.4507	0.0000
Slope	-0.0592037	0.00472156	-12.539	0.0000

Analysis of Variance

Source	*Sum of Squares*	*Df*	*Mean Square*	*F-Ratio*	*P-Value*
Model	1709.94	1	1709.94	157.23	0.0000
Residual	2936.42	270	10.8756		
Total (Corr.)	4646.35	271			

Correlation Coefficient = -0.606644
R-squared = 36.8017 percent
R-squared (adjusted for d.f.) = 36.5676 percent
Standard Error of Est. = 3.29782
Mean absolute error = 2.64739
Durbin-Watson statistic = 1.67522 (P=0.0036)
Lag 1 residual autocorrelation = 0.159998

The scatterplot of the data is also shown below.



a. Determine the explanatory variable, response variable, and list any lurking variables
b. Identify and interpret the correlation coefficient according to the 3 metrics that we discussed
c. Write out the equation of the regression line that fits the model
d. Interpret the slope for the regression line in question.
e. For an individual who weight 224 pounds, what is their predicted vertical leap, in inches?
f. Create a 95% confidence interval for the prediction for part e.
g. Write out the null and alternative hypothesis and either reject or accept the null? Complete and write out all steps of the hypothesis test and write a concluding statement.

109. Following this test, during the 2019 NFL combine, the evaluators wanted to test the relationship between the weight of individuals and their ability to complete bench press repetitions (as a measure of their total strength). The number of repetitions they completed was measured and the following regression output was obtained.

Simple Regression - Bench_Press_Reps vs. Weight_3_lbs

Dependent variable: Bench_Press_Reps
Independent variable: Weight_3_lbs
Linear model: Y = a + b*X
Number of observations: 263

Coefficients

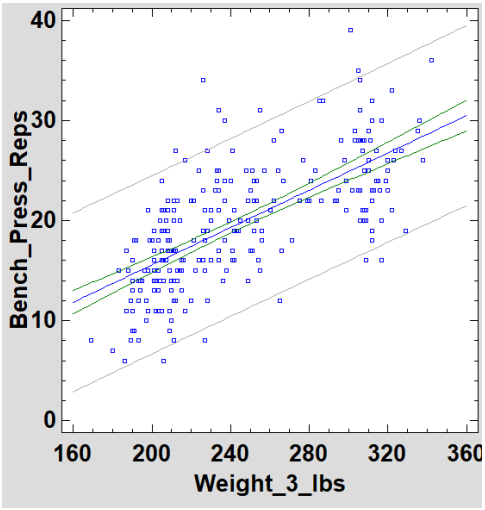
	Least Squares	*Standard*	*T*	
Parameter	*Estimate*	*Error*	*Statistic*	*P-Value*
Intercept	-3.06495	1.5575	-1.96786	0.0501
Slope	0.0931959	0.00627828	14.8442	0.0000

Analysis of Variance

Source	*Sum of Squares*	*Df*	*Mean Square*	*F-Ratio*	*P-Value*
Model	4469.16	1	4469.16	220.35	0.0000
Residual	5293.64	261	20.2822		
Total (Corr.)	9762.81	262			

Correlation Coefficient = 0.67659
R-squared = 45.7775 percent
R-squared (adjusted for d.f.) = 45.5697 percent
Standard Error of Est. = 4.50357
Mean absolute error = 3.56103
Durbin-Watson statistic = 1.87196 (P=0.1500)
Lag 1 residual autocorrelation = 0.0527008

The scatterplot of the data is also shown below.



a. Determine the explanatory variable, response variable, and list any lurking variables
b. Identify and interpret the correlation coefficient according to the 3 metrics that we discussed
c. Write out the equation of the regression line that fits the model
d. Interpret the slope for the regression line in question.
e. For an individual who weight 234 pounds, what is their predicted number of bench press reps that they can complete?
f. Create a 95% confidence interval for the prediction for part e.
g. Write out the null and alternative hypothesis and either reject or accept the null? Complete and write out all steps of the hypothesis test and write a concluding statement.

110. As a final test at the 2019 NFL combine, the evaluators wanted to test the relationship between the weight of individuals and the length that they were able to jump on the broad jump, in inches. The following regression output was obtained.

Simple Regression - Broad_Jump_Inches vs. Weight_4_lbs

Dependent variable: Broad_Jump_Inches
Independent variable: Weight_4_lbs
Linear model: Y = a + b*X
Number of observations: 271

Coefficients

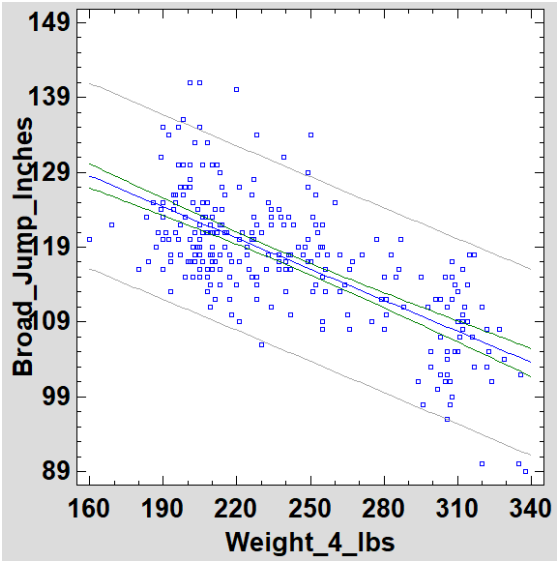
	Least Squares	*Standard*	*T*	
Parameter	*Estimate*	*Error*	*Statistic*	*P-Value*
Intercept	150.61	2.16177	69.6699	0.0000
Slope	-0.138247	0.00882851	-15.6592	0.0000

Analysis of Variance

Source	*Sum of Squares*	*Df*	*Mean Square*	*F-Ratio*	*P-Value*
Model	9562.67	1	9562.67	245.21	0.0000
Residual	10490.5	269	38.998		
Total (Corr.)	20053.1	270			

Correlation Coefficient = -0.690555
R-squared = 47.6867 percent
R-squared (adjusted for d.f.) = 47.4922 percent
Standard Error of Est. = 6.24484
Mean absolute error = 4.90182
Durbin-Watson statistic = 1.91282 (P=0.2370)
Lag 1 residual autocorrelation = 0.0409305

The scatterplot of the data is also shown below.



a. Determine the explanatory variable, response variable, and list any lurking variables
b. Identify and interpret the correlation coefficient according to the 3 metrics that we discussed
c. Write out the equation of the regression line that fits the model
d. Interpret the slope for the regression line in question.
e. For an individual who weighed 237 pounds, what is their predicted number of inches that they can jump for the broad jump?
f. Create a 95% confidence interval for the prediction for part e.
g. Write out the null and alternative hypothesis and either reject or accept the null? Complete and write out all steps of the hypothesis test and write a concluding statement.

111. For those who want to watch their weight and follow a nutritious diet, it isn't difficult to find guidelines and recommendations. That said, there is subjectivity. One of the questions that researchers have asked is whether Americans in the general public are good at identifying healthy foods. To be specific, are American adults good ag identifying foods that would be considered to be healthy when compared to health experts? In 2016, a survey was conducted on this very fact. A regression was done to compare the percentage of

nutritionists who would say a certain food is healthy, versus the percentage of American adults who would say the same item is health. A sample table is noted here.

Food	Nutritionists_ Score	Public_Score
granola bar	28	71
coconut oil	37	72
frozen yogurt	32	66
granola	47	80
slim fast shake	21	47
orange juice	62	78
american cheese	24	39
quinoa	89	58
tofu	85	57
sushi	75	49
hummus	90	66
wine	70	52
shrimp	85	69
popcorn	61	52
pork chops	59	52
whole milk	63	59
steak	60	63
cheddar cheese	57	56
hamburgers	28	29
beef jerky	23	27
diet soda	18	16
white bread	15	18
chocolate chip	6	10
apples	99	96
oranges	99	96
oatmeal	97	92
chicken	91	91
turkey	91	90
peanut butter	81	79
baked potatoes	72	71
almonds	98	92
avocados	95	86
bacon	12	22
beer	23	13
butter	40	35
canned tuna	84	70

The following regression output was obtained for the data in question.

Simple Regression - Public_Score vs. Nutritionists_Score

Dependent variable: Public_Score
Independent variable: Nutritionists_Score
Linear model: Y = a + b*X
Number of observations: 52

Coefficients

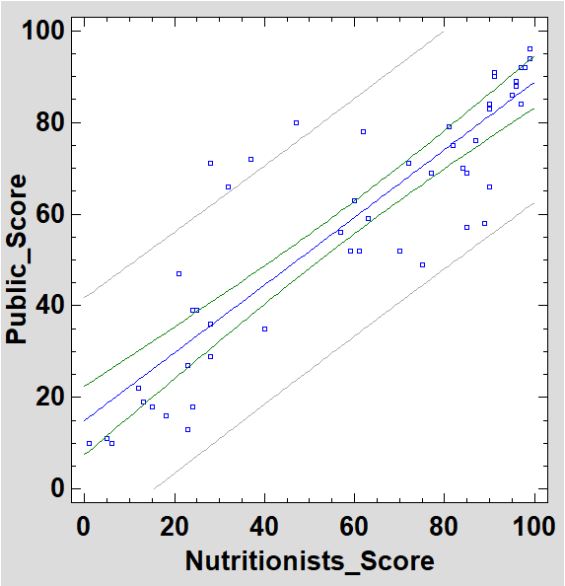
	Least Squares	*Standard*	*T*	
Parameter	*Estimate*	*Error*	*Statistic*	*P-Value*
Intercept	15.0035	3.73847	4.01328	0.0002
Slope	0.738265	0.0547465	13.4852	0.0000

Analysis of Variance

Source	*Sum of Squares*	*Df*	*Mean Square*	*F-Ratio*	*P-Value*
Model	29737.8	1	29737.8	181.85	0.0000
Residual	8176.49	50	163.53		
Total (Corr.)	37914.3	51			

Correlation Coefficient = 0.885631
R-squared = 78.4343 percent
R-squared (adjusted for d.f.) = 78.003 percent
Standard Error of Est. = 12.7879
Mean absolute error = 9.37616
Durbin-Watson statistic = 0.379706 (P=0.0000)
Lag 1 residual autocorrelation = 0.733441

The following scatterplot was also obtained:



a. Determine the explanatory variable, response variable, and list any lurking variables
b. Identify and interpret the correlation coefficient.
c. Write out the equation of the regression line that fits the model.
d. Interpret the slope for the regression line in question.
e. For an food that scores 71 on the nutritionists axis (aka 71% of nutritionists agreed that it is nutritious), what is the predicted public score (aka the percentage of individuals in the public who agree that the food is nutritious?)
f. Create a 95% confidence interval for the prediction for part e.
g. Write out the null and alternative hypothesis and either reject or accept the null? Complete and write out all steps of the hypothesis test and write a concluding statement.

112. In 2016, the World Bank published a report looking at the relationship between the gross domestic product (GDP) per capita of many countries, and their life expectancy (in years). GDP is the total value of goods and services that are produced in a country, converted into dollars. We might expect people in rich countries to

live longer, though there is a question of whether the data is linear. The regression output for the regressions between the two quantitative variables is shown below, as is the scatterplot.

Simple Regression - Life_Expectancy vs. GDP

Dependent variable: Life_Expectancy
Independent variable: GDP
Linear model: Y = a + b*X
Number of observations: 234

Coefficients

	Least Squares	*Standard*	*T*	
Parameter	*Estimate*	*Error*	*Statistic*	*P-Value*
Intercept	68.4532	0.478768	142.978	0.0000
Slope	0.000235154	0.0000198399	11.8526	0.0000

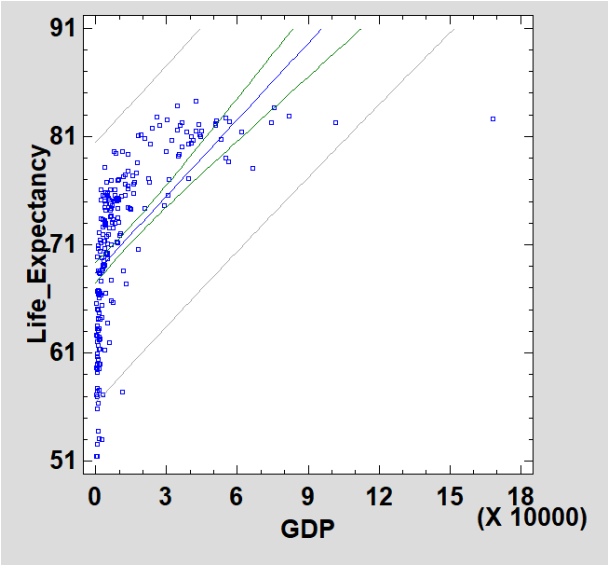
Analysis of Variance

Source	*Sum of Squares*	*Df*	*Mean Square*	*F-Ratio*	*P-Value*
Model	5216.18	1	5216.18	140.48	0.0000
Residual	8614.16	232	37.13		
Total (Corr.)	13830.3	233			

Correlation Coefficient = 0.614129
R-squared = 37.7155 percent
R-squared (adjusted for d.f.) = 37.447 percent
Standard Error of Est. = 6.09344
Mean absolute error = 4.86502
Durbin-Watson statistic = 1.99663 (P=0.4897)
Lag 1 residual autocorrelation = -0.0039411

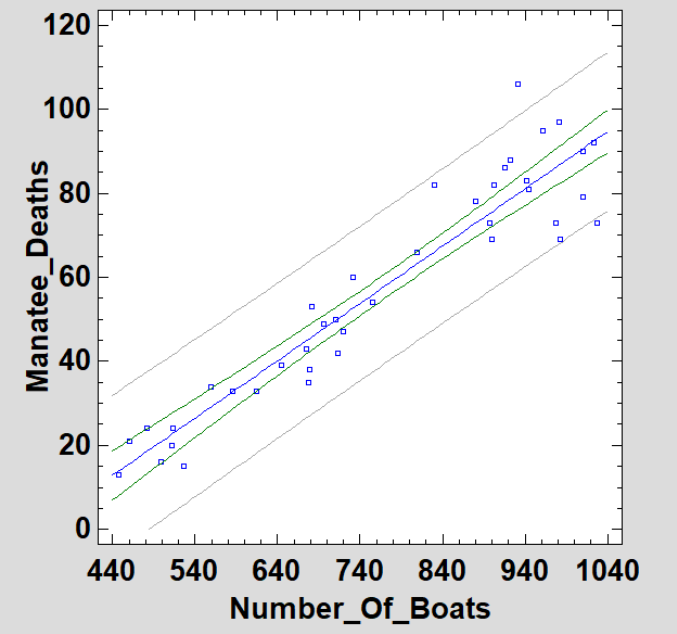
Here is the scatterplot for the data.

a. Determine the explanatory variable, response variable, and list any lurking variables
b. Identify and interpret the correlation coefficient.
c. Write out the equation of the regression line that fits the model.
d. Interpret the slope for the regression line in question.
e. For a country that has a GDP of 30,000, what is the predicted life expectancy in years?
f. Create a 95% confidence interval for the prediction for part e.
g. Write out the null and alternative hypothesis and either reject or accept the null? Complete and write out all steps of the hypothesis test and write a concluding statement.



113. Manatees are large, gentle, slow-moving creatures found along the coast of Florida. Many manatees are injured or killed by boats. The following scatterplot shows the relationship between the number of manatee

deaths by boats versus the number of boats registered in Florida (in thousands) for the years between 1977 and 2016.



The following is the regression output obtained looking at the relationship between the number of boats and the number of manatee deaths.

Simple Regression - Manatee_Deaths vs. Number_Of_Boats

Dependent variable: Manatee_Deaths
Independent variable: Number_Of_Boats
Linear model: Y = a + b*X
Number of observations: 40

Coefficients

	Least Squares	*Standard*	*T*	
Parameter	*Estimate*	*Error*	*Statistic*	*P-Value*
Intercept	-47.1559	6.13059	-7.6919	0.0000
Slope	0.136398	0.00776426	17.5674	0.0000

Analysis of Variance

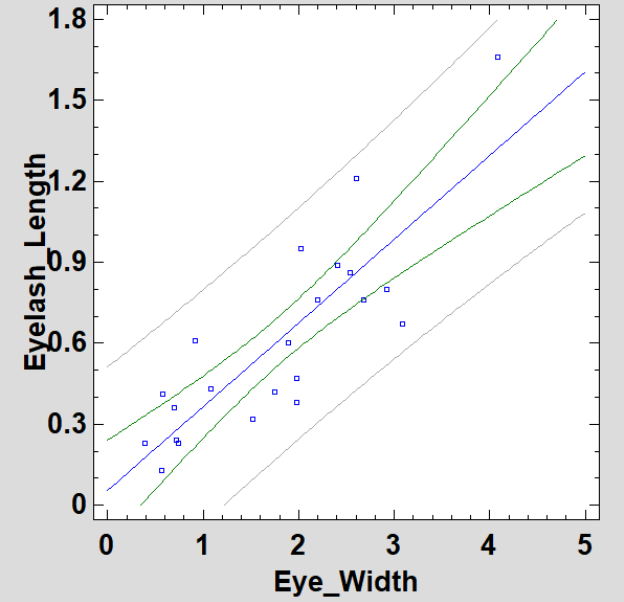
Source	*Sum of Squares*	*Df*	*Mean Square*	*F-Ratio*	*P-Value*
Model	24798.0	1	24798.0	308.61	0.0000
Residual	3053.41	38	80.3528		
Total (Corr.)	27851.4	39			

Correlation Coefficient = 0.943593
R-squared = 89.0368 percent
R-squared (adjusted for d.f.) = 88.7483 percent
Standard Error of Est. = 8.96397
Mean absolute error = 6.50052
Durbin-Watson statistic = 1.40601 (P=0.0175)
Lag 1 residual autocorrelation = 0.184742

a. Determine the explanatory variable, response variable, and list any lurking variables
b. Identify and interpret the correlation coefficient.

c. Write out the equation of the regression line that fits the model.
d. Interpret the slope for the regression line in question.
e. For a year in which 766 thousand boats were registered in Florida (in thousands), what is the predicted number of manatee deaths?
f. Create a 95% confidence interval for the prediction for part e.
g. Write out the null and alternative hypothesis and either reject or accept the null? Complete and write out all steps of the hypothesis test and write a concluding statement.

114. False eyelashes are sometimes worn in an effort to extend the length of an individual's natural eyelashes. Research suggests, however, that our natural eyelash length is proportional to the width of our eye. To better understand how eye width length (in centimeters or cm), is related to the eyelash length (in centimeters, or cm), measurements were obtained from 22 mammals. The scatterplot for this data was noted below.



Regression output for the data was also obtained and is shown below.

Simple Regression - Eyelash_Length vs. Eye_Width

Dependent variable: Eyelash_Length
Independent variable: Eye_Width
Linear model: Y = a + b*X
Number of observations: 22

Coefficients

	Least Squares	*Standard*	*T*	
Parameter	*Estimate*	*Error*	*Statistic*	*P-Value*
Intercept	0.053748	0.0900085	0.597143	0.5571
Slope	0.310151	0.0442534	7.00852	0.0000

Analysis of Variance

Source	*Sum of Squares*	*Df*	*Mean Square*	*F-Ratio*	*P-Value*
Model	1.9809	1	1.9809	49.12	0.0000
Residual	0.806564	20	0.0403282		
Total (Corr.)	2.78746	21			

Correlation Coefficient = 0.842998
R-squared = 71.0645 percent
R-squared (adjusted for d.f.) = 69.6178 percent
Standard Error of Est. = 0.200819
Mean absolute error = 0.156784
Durbin-Watson statistic = 1.84412 (P=0.3136)
Lag 1 residual autocorrelation = -0.000881128

a. Determine the explanatory variable, response variable, and list any lurking variables
b. Identify and interpret the correlation coefficient.

c. Write out the equation of the regression line that fits the model.
d. Interpret the slope for the regression line in question.
e. For a mammal with an eye width (cm) of 2.41 (a snow leopard), what is the predicted eyelash length (cm). How does this compare to the actual value of eyelash length (cm) for snow leopards are 0.89.
f. Create a 95% confidence interval for the prediction for part e.
g. Write out the null and alternative hypothesis and either reject or accept the null? Complete and write out all steps of the hypothesis test and write a concluding statement.

115. A recent study looked at the relationship between the price charged for a hot dog at a Major League Baseball game and the price charged for an ounce of beer. The regression and scatterplot from this analysis are shown below.

Simple Regression - Beer_Average_Price vs. Hot_Dog_Avg_Price

Dependent variable: Beer_Average_Price
Independent variable: Hot_Dog_Avg_Price
Linear model: Y = a + b*X
Number of observations: 24

Coefficients

	Least Squares	*Standard*	*T*	
Parameter	*Estimate*	*Error*	*Statistic*	*P-Value*
Intercept	0.319382	0.116798	2.73449	0.0121
Slope	0.0223054	0.02265	0.984783	0.3354

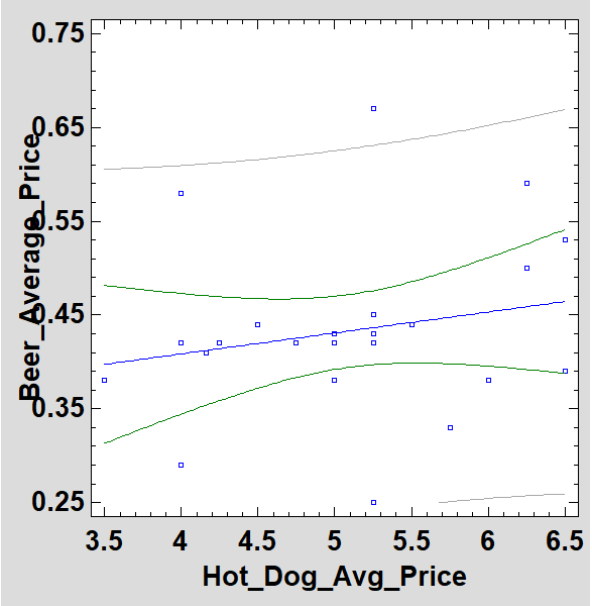
Analysis of Variance

Source	*Sum of Squares*	*Df*	*Mean Square*	*F-Ratio*	*P-Value*
Model	0.00815261	1	0.00815261	0.97	0.3354
Residual	0.184943	22	0.00840651		
Total (Corr.)	0.193096	23			

Correlation Coefficient = 0.205476
R-squared = 4.22205 percent
R-squared (adjusted for d.f.) = -0.131491 percent
Standard Error of Est. = 0.091687
Mean absolute error = 0.057828
Durbin-Watson statistic = 2.17238 (P=0.6639)
Lag 1 residual autocorrelation = -0.0943054

The scatterplot is also shown below.

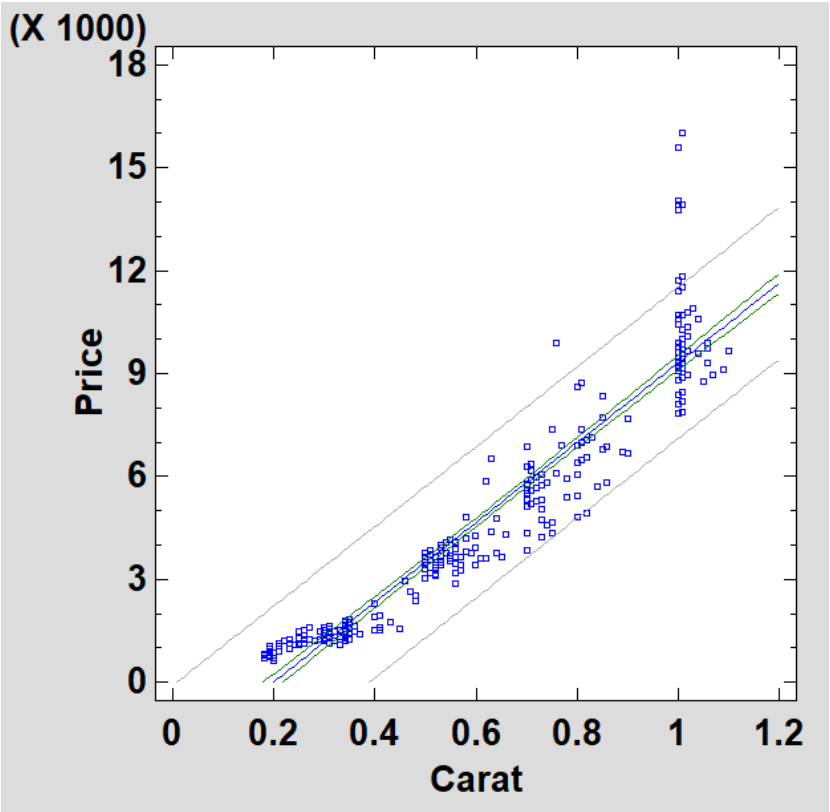
a. Determine the explanatory variable, response variable, and list any lurking variables that you think may exist for this problem? Why do you think they are lurking variables?
b. Identify and interpret the correlation coefficient.
c. Write out the equation of the regression line that fits the model.
d. Interpret the slope for the regression line in question.
e. For a stadium / MLB team that has a hot dog price of 5.00, what is the predicted price of beer? Note that for an actual measurement, the Angels stadium has a price of a Hot Dog of $5.00 with an actual provide of Beer per ounce of $0.38.
f. Create a 95% confidence interval for the prediction for part e.



g. Write out the null and alternative hypothesis and either reject or accept the null? Complete and write out all steps of the hypothesis test and write a concluding statement.

116. A study in South Korea recently looked at the relationship between the price of a diamond and its many attributes. The following scatterplot was obtained, showing the price of a diamond versus the number of carats that the diamond had. The total number of observations within the study was 308.

When looking at the scatterplot, what do you notice about the shape of the scatterplot. Are any regression assumptions violated? Which ones concern you and why? What are your options for dealing with these violations?



The regression analysis of the data is shown below. For this analysis, complete the relevant questions.

Simple Regression - Price vs. Carat

Dependent variable: Price
Independent variable: Carat
Linear model: Y = a + b*X
Number of observations: 308

Coefficients

	Least Squares	*Standard*	*T*	
Parameter	*Estimate*	*Error*	*Statistic*	*P-Value*
Intercept	-2298.36	158.531	-14.4979	0.0000
Slope	11598.9	230.111	50.4057	0.0000

Analysis of Variance

Source	*Sum of Squares*	*Df*	*Mean Square*	*F-Ratio*	*P-Value*
Model	3.17325E9	1	3.17325E9	2540.73	0.0000
Residual	3.82179E8	306	1.24895E6		
Total (Corr.)	3.55543E9	307			

Correlation Coefficient = 0.944727
R-squared = 89.2508 percent
R-squared (adjusted for d.f.) = 89.2157 percent
Standard Error of Est. = 1117.56
Mean absolute error = 742.994
Durbin-Watson statistic = 1.21563 (P=0.0000)
Lag 1 residual autocorrelation = 0.390163

a. Determine the explanatory variable, response variable, and list any lurking variables that you think may exist for this problem? Why do you think they are lurking variables?
b. Identify and interpret the correlation coefficient.
c. Write out the equation of the regression line that fits the model.
d. Interpret the slope for the regression line in question.
e. For a diamond that has a carat of 0.7, what is the predicted price?
f. Create a 95% confidence interval for the prediction for part e.
g. Write out the null and alternative hypothesis and either reject or accept the null? Complete and write out all steps of the hypothesis test and write a concluding statement

117. A study in transportation safety collected data on 42 North American cities. From each city, two of the variables recorded were X = percentage of licensed drivers who are under 21 years of age, and Y = the number of fatal accidents per year per 1000 licenses. Below is the output from the data:

Parameter	**Std. Estimate**	**Error**	**T Statistic**	**p-value**
Intercept	-1.59741	0.371671	-4.29792	0.0001
Slope	0.287053	0.0293898	9.76711	Unknown

Correlation Coefficient = 0.839387
R-squared = 70.4571 percent
Standard error of estimate = 0.58935

a) What is the formula for the regression function?
b) Interpret the slope of the regression equation.
c) Construct an 95% interval to predict y when x = 14, and explain what it means in the context of the problem.
d) Advanced question? What is the p value for the slope in this problem? Calculate it and show your work?
e) Based on this analysis, can we conclude that the number of fatal accidents is linearly related to the percentage of licensed drives? Justify your answer showing all work. What is the null and the alternative hypothesis?

118. A study was conducted to understand the relationship between the debt that a family has and the number of hours that they spend watching television each week on average. Below are the results that were obtained within StatGraphics.

Dependent variable: Debt
Independent variable: Television

Parameter	Estimate	Std. Error	T-Statistic	P-Value
Intercept	48039.7	6004.01	8.00127	0.0000
Slope	2581.79	187.545	13.7662	0.0000

Analysis of Variance

Source	Sum of Squares	Df	Mean Square	F-Ratio	P-Value
Model	2.83406E11	1	2.83406E11	189.51	0.0000
Residual	6.40061E11	428	1.49547E9		
Total(Corr.)	9.23467E11	429			

Correlation Coefficient = 0.55398
R-squared = 30.6893 percent
Standard Error of Est. = 38671.3

a) What is the formula for the regression function?
b) Interpret the slope of the regression equation.
c) Construct an 95% interval to predict the debt for a family that watches 20 hours of TV on average per week.
d) Based on this analysis, can we conclude that the amount of debt a family has is related to the number of hours of television watched. Justify your answer showing all work. What is the null and the alternative hypothesis?

119. A study was conducted showing the relationship between the average maintenance cost and the age of a car in years. Below is the output from StatGraphics.

Simple Regression - Maintenance Cost vs. Age In Years

Dependent variable: Maintenance Cost
Independent variable: Age In Years
Linear model: Y = a + b*X
Number of observations: 13

Coefficients

	Least Squares	*Standard*	*T*	
Parameter	*Estimate*	*Error*	*Statistic*	*P-Value*
Intercept	54.7757	54.87	0.998282	0.3396
Slope	120.689	11.8442	10.1898	0.0000

Analysis of Variance

Source	*Sum of Squares*	*Df*	*Mean Square*	*F-Ratio*	*P-Value*
Model	674514.	1	674514.	103.83	0.0000
Residual	71458.8	11	6496.25		
Total (Corr.)	745973.	12			

Correlation Coefficient = 0.950898
R-squared = 90.4207 percent
R-squared (adjusted for d.f.) = 89.5499 percent
Standard Error of Est. = 80.5993
Mean absolute error = 64.7521
Durbin-Watson statistic = 1.24988 (P=0.0338)
Lag 1 residual autocorrelation = 0.317439

a) What is the form, direction and strength of the correlation?
b) What is the formula for the regression function based on the output above?
c) Interpret the slope of the regression equation?
d) Construct an 95% interval to predict the maintenance cost for a car that is 7 years old?
e) Based on this analysis, can we conclude that a relationship exists between the maintenance cost and the age of the car in years? What is the null and alternative hypothesis? Justify your answer showing all work.

120. The following shows the results of a study that shows the relationship between the average gross income a person makes and the charitable contributions that they make in dollars. The following is StatGraphics output.

Simple Regression - Charitable Contributions vs. Average Gross Income

Dependent variable: Charitable Contributions
Independent variable: Average Gross Income
Linear model: Y = a + b*X
Number of observations: 11

Coefficients

	Least Squares	*Standard*	*T*	
Parameter	*Estimate*	*Error*	*Statistic*	*P-Value*
Intercept	-6169.94	1510.36	-4.08507	0.0027
Slope	195.698	18.9155	10.3459	0.0000

Analysis of Variance

Source	*Sum of Squares*	*Df*	*Mean Square*	*F-Ratio*	*P-Value*
Model	1.04943E8	1	1.04943E8	107.04	0.0000
Residual	8.8238E6	9	980423.		
Total (Corr.)	1.13767E8	10			

Correlation Coefficient = 0.960437
R-squared = 92.244 percent
R-squared (adjusted for d.f.) = 91.3822 percent
Standard Error of Est. = 990.163
Mean absolute error = 683.393
Durbin-Watson statistic = 2.66133 (P=0.7900)
Lag 1 residual autocorrelation = -0.344421

a) What is the form, direction and strength of the correlation?
b) What is the formula for the regression function based on the output above?
c) Interpret the slope of the regression equation?
d) Construct an 95% interval to predict the charitable contributions that a person would make given that they have an annual income of $57,000? Note that the regression was done with the data already in thousands?
e) Based on this analysis, can we conclude that a relationship exists between charitable contributions and the average gross income? What is the null and alternative hypothesis? Justify your answer showing all work.

121. The following data shows the relationship between the first exam (first test) and a students final grade. The following are results from StatGraphics.

Simple Regression - Final Test Grade vs. 1st Test Grade

Dependent variable: Final Test Grade
Independent variable: 1st Test Grade
Linear model: Y = a + b*X
Number of observations: 9

Coefficients

	Least Squares	*Standard*	*T*	
Parameter	*Estimate*	*Error*	*Statistic*	*P-Value*
Intercept	101.435	26.3263	3.85301	0.0063
Slope	-0.280711	0.320417	-0.876079	0.4100

Analysis of Variance

Source	*Sum of Squares*	*Df*	*Mean Square*	*F-Ratio*	*P-Value*
Model	121.735	1	121.735	0.77	0.4100
Residual	1110.27	7	158.609		
Total (Corr.)	1232.0	8			

Correlation Coefficient = -0.314342
R-squared = 9.88107 percent
R-squared (adjusted for d.f.) = -2.99306 percent
Standard Error of Est. = 12.594
Mean absolute error = 9.69635
Durbin-Watson statistic = 2.19484 (P=0.6542)
Lag 1 residual autocorrelation = -0.118248

a) What is the form, direction and strength of the correlation?
b) What is the formula for the regression function based on the output above?
c) Interpret the slope of the regression equation?
d) Construct an 95% interval to predict the final grade a person would get if they scored an 87 on their first midterm?
e) Based on this analysis, can we conclude that a relationship exists between charitable contributions and the average gross income? What is the null and alternative hypothesis? Justify your answer showing all work.

122. The following data shows the results from a regression that was run on a group of companies, estimating the relationship between long term assets and long term debt.

Simple Regression - Long Term Debt vs. Long Term Assets

Dependent variable: Long Term Debt
Independent variable: Long Term Assets
Linear model: Y = a + b*X
Number of observations: 12

Coefficients

	Least Squares	*Standard*	*T*	
Parameter	*Estimate*	*Error*	*Statistic*	*P-Value*
Intercept	-7.84048	5.68931	-1.37811	0.1982
Slope	0.812132	0.108006	7.51933	0.0000

Analysis of Variance

Source	*Sum of Squares*	*Df*	*Mean Square*	*F-Ratio*	*P-Value*
Model	1037.43	1	1037.43	56.54	0.0000
Residual	183.485	10	18.3485		
Total (Corr.)	1220.92	11			

Correlation Coefficient = 0.9218
R-squared = 84.9715 percent
R-squared (adjusted for d.f.) = 83.4687 percent
Standard Error of Est. = 4.28352
Mean absolute error = 3.34034
Durbin-Watson statistic = 1.11765 (P=0.0201)
Lag 1 residual autocorrelation = 0.420836

a) What is the form, direction and strength of the correlation?
b) What is the formula for the regression function based on the output above?
c) Interpret the slope of the regression equation?
d) Construct an 95% interval to predict the amount of debt that a company has if they had assets of 100 million? Note that the regression assumed that the data was already in millions?
e) Based on this analysis, can we conclude that a relationship exists between long term debt and long term assets? What is the null and alternative hypothesis? Justify your answer showing all work.

Discrete Probability Distributions (Midterm 2)

123. In Gallup's Annual consumption Habits poll, telephone interviews were conducted for a random sample of 1014 adults aged 18 and over. One question was how many cups of coffee to you drink on average? The following table shows the results that were obtained.

Number Of Cups Per Day	0	1	2	3	4 or more
Observations	365	264	193	91	101

Develop a random variable x = number of cups of coffee consumed on an average day.

a) Use the relative frequency approach to develop a probability distribution for x

b) Compute the expected value for x and interpret this value based on the data provided?
c) Let us assume that individuals spend an average of $1.22 per cup of coffee. Develop a new random variable that represents the average amount spent per person on coffee each day relating to the number of cups of coffee that they have consumed. For this new random variable, create a new probability distribution for it and calculate the expected value for the number of dollars spent on average per day on coffee? You can also calculate the variance of this random variable if you would like.

124. Consider a play of the casino game `Quick Draw'. In this game, a player pays $9 to play. The player picks one card from a standard pack of 52 cards (i.e. there are four A's and four K's in a standard pack of 52 cards). If the player gets an Ace, they win $49 (i.e. the profit is $40); if the player selects a King, they win $29 (i.e. the profit is $20). Otherwise, the player wins nothing and also loses the bet of $9. Let the random variable X represent the player's profit on a single play of `Quick Draw'.
   a. Construct the probability distribution for the random variable X.
   b. Calculate the expected value of the random variable X and provide an interpretation of its value, in context of this question.

125. On any given day, a car dealership sells 0, 1, 2, 3, or 4 cars with probabilities 0.05, 0.20, 0.25, 0.40, 0.10 respectively. (They never sell more than 4 cars). Suppose they make a profit of $3500 from each car sold, but they have a fixed cost of $1700 per day for rent and salaries. Let the random variable X = the net profit per day. (i.e., the profit from the sales of cars minus fixed cost).
   a. Construct the probability distribution table for x
   b. Calculate the expected value (or mean) of this probability distribution, and explain what the number means in the context of the problem.

126. Line management at Starbucks is an important tool to both predict revenue and determine the optimal staffing levels. Imagine that we watch the line at Starbucks for 60 minutes and recorded the number of individuals in line for each minute (n = 60). The data was tallied together and the following statistics were obtained.

Number Of People In Line	0	1	2	3	4	5	6
Observations	4	12	18	10	11	3	2

a) Use the relative frequency approach to construct an empirical discrete probability distributions for the number of people in line on average each minute at a particular Starbucks location
b) Draw a graph of this probability distribution of this random variable
c) Show that your probability distribution satisfies the required conditions for a valid discrete probability distribution
d) Calculate the expected value of this random variable and interpret it. What is the average number of people that you expect to be in line during a given minute?
e) Imagine that we change the random variable to reflect the average revenue obtained each minute at a random Starbucks location. In particular, at a random location, let us assume that the mean revenue per drink is $3.74. Redo the question, and create a random variable X which reflect the average revenue per minute at a given location. Create the discrete probability distribution, draw the graph showing the PDF of this random variable, and calculate the expected value for this random variable.

127. Line management at Starbucks is an important tool to both predict revenue and determine the optimal staffing levels. Imagine that the number of staff that are brought on each minute changes depending on the number of people in line. In particular, if there are 0 – 2 people in line, 1 staff member is put on shift. If there are

3 – 5 people in line, 2 people are on the floor. Finally, if there are 6 people in line, 3 people are asked to work. Imagine that we watch the line at Starbucks for 60 minutes and recorded the number of individuals in line for each minute (n = 60). The data was tallied together and the following statistics were obtained.

Number Of People In Line	0	1	2	3	4	5	6
Observations	4	12	18	10	11	3	2

f) Use the relative frequency approach to construct an empirical discrete probability distributions for the random variable corresponding to the number of staff members that should be on shift during a given minute at a particular Starbucks location.
g) Draw a graph of this probability distribution of this random variable
h) Show that your probability distribution satisfies the required conditions for a valid discrete probability distribution
i) Calculate the expected value of this random variable and interpret it. What is the average number of people that you expect to be in line during a given minute?

128. Equipment management within a fitness center is an important problem. Too often treadmills are full when individuals come, especially during peak time. To better understand this, a fitness center set out to monitor the number of treadmills in use during a particular hour of the day. In particular, for every minute during a single hour, they recorded the number of treadmills that were used.

Number Of Treadmills Used	0	1	2	3	4	5	6
Count Of Minutes Usage Observed For The Treadmills	3	6	14	20	9	4	2

a) Use the relative frequency approach to construct an empirical discrete probability distributions for the random variable corresponding to the number of treadmills that are in use during a given minute at a particular fitness center.
b) Draw a graph of this probability distribution of this random variable
c) Show that your probability distribution satisfies the required conditions for a valid discrete probability distribution
d) Calculate the expected value of this random variable and interpret it. What is the average number of treadmills that you expect to see used during a given minute?

129. An insurance company offers a disability policy that will pay workers $15,000 if they become unable to work for 1 to 6 months, and $40,000 if they become unable to work for more than 6 months. (The company does not pay workers if they do not become unable to work for at least 1 month). It costs a worker $500 to enrol. Based on past data, the company found that there is a 0.015 (1.5%) chance that a worker will become unable to work for 1 to 6 months, and 0.005 (0.5%) chance they become unable to work for more than 6 months. Define X as the amount of money the insurance company will make on a policy. (i.e., the profit from each policy from the point of view of the insurance company.)

a. Develop the Probability Distribution for X. [3 marks]
b. Calculate the expected value of X. And explain what it means in the context of the problem. [3 marks]

130. Consider the experiment of tossing a coin twice.
a) List the experimental outcomes
b) Define a random variable that represents the number of heads occurring on the two tosses
c) Show what value the random variable would assume for each of the experimental outcomes?

d) Is this random variable continuous or discrete?

131. The following data were collected by counting the number of operating rooms in use at Tampa General Hospital over a 20-day period.

- On 3 days only one of the operating rooms was used
- On five days two were used
- On eight days three were used
- On four days all four of the hospitals operating rooms were used

e) Use the relative frequency approach to construct an empirical discrete probability distributions for the number of operating rooms used on any given day
f) Draw a graph of this probability distribution
g) Show that your probability distribution satisfies the required conditions for a valid discrete probability distribution

132. On any given day, a rental dealer rents 0, 1, 2, 3, or 4 cars with probabilities 0.10, 0.15, 0.25, 0.30, 0.20 respectively. (They never rent more than 4 cars). Suppose they make a profit of $2100 from each car rented, but they have a fixed cost of $1400 per day for rent and salaries at the dealer. Let the random variable X = the net profit per day. (i.e., the profit from the rent of cars minus the fixed cost).

a) Develop a probability distribution for the net profit per day
b) Draw a graph of the probability distribution
c) Show that your probability distribution satisfies the conditions for a discrete probability distribution.
d) Calculate the expected value of the probability distribution

133. In the Arizona lottery called Pick 3, a player pays $1 and then picks a three-digit number. If those three numbers are picked in that specific order the person wins $500.

a. Create the probability distribution for this particular game? What are the X's and f(x) values?
b. What is the expected value in this game?

134. The Ohio lottery has a game called Pick 4 where a player pays $1 and picks a four-digit number. If the four numbers come up in the order you picked, then you win $2,500. What is your expected value?

a. Create the probability distribution for this particular game? What are the X's and f(x) values?
b. What is the expected value in this game?

135. The New York city police department tracks the number of vehicles involved in each vehicle collision in Manhattan. The table shows the frequency distribution for X, which is the number of vehicles involved in collisions in Manhattan in July 2014.

X = Vehicles Involved	Frequency
1	288
2	3151
3	109
4	12
5	4
6	0
7	0
8	1

a) Create a probability distribution for the random variable in question?
b) Draw a graph of the probability distribution for the random variable in question?
c) What is the probability that at least 3 vehicles will be involved in a collision?
d) What is the probability that 6 vehicles will be involved in a collision?
e) What is the expected value for the number of vehicles that will be involved in collisions in crashes on a given day in July?

136. The New York city police department tracks the number of people injured in each collision that occurs within Manhattan. The table shows the frequency distribution for X, which is the number of people injured in collisions that occur in Manhattan in July 2014.

X = number of injured	Frequency
0	3028
1	462
2	57
3	10
4	3
5	2
6	1
7	2

a) Create a probability distribution for the random variable in question?
b) Draw a graph of the probability distribution for the random variable in question?
c) What is the probability that at least 2 people will be injured in a collision?
d) What is the probability that 6 or more people will be injured in a collision?
e) What is the expected value for the number of people who will be injured in collisions in crashes on a given day in July?

137. The census bureau in the US conducted a survey to determine the distribution for the number of people in an American household. After completing the survey, the following raw data was obtained.

Family Size	1	2	3	4	5	6	7
Count	281	352	151	132	63	21	11

a) Create a probability distribution for the random variable in this case. What is the random variable in question?
b) Draw a graph showing the probability distribution for the random variable in this question?
c) What is the expected value for the average family size in this particular case?

138. The description of grades for a course that was considered an easy statistics course in a random university was provided below. Suppose that, after surveying students, the following distribution was obtained.

Grade	A	B	C	D	F
Count	41	32	21	11	2

a) Create a probability distribution for the random variable in this case. What is the random variable in question?
b) Draw a graph showing the probability distribution for the random variable in this question?
c) Let us assume that A's get a 4, B's a 3, C's a 2, D's a 1, and F's a 0. What is the expected grade for a random student based on the probability distribution.

139. The percent frequency distributions of job satisfaction scores for a sample of information systems executives and managers are shown. The scores range from 1 (very dissatisfied) to 5 (very satisfied).

Job Satisfaction Score	IS Senior Executive (%)	IS Middle Managers (%)

1	5	4
2	9	10
3	3	12
4	42	46
5	41	28

a) Develop a probability distribution for the job satisfaction score of a senior executive
b) Develop a probability distribution for the job satisfaction score of a middle manager
c) What is the probability a senior executive will report a job satisfaction score of 4 or 5?
d) What is the probability a middle manager is very satisfied?
e) Compare the overall job satisfaction of senior executives and middle managers?

140. An insurance company offers farmers a policy to protect their crops. In this case, the policy will pay $100,000 if there are between 20 and 40 days of no rain (drought) during the growing season on their farms (causing floods), and $300,000 if there are more than 40 days of no rain rain (drought) on their farms causing massive plant failure. It costs farmers $5000 to enrol and get this valuable crop insurance. Based on past data, the company found that in any given year there is a 0.02 (2%) chance that there will be between 20 and 40 days of no rain during the growing season, and a 0.005 (0.5%) chance that there will be more than 40 days of no rain during the growing season. Define X as the amount of money the insurance company will make on a policy. (i.e., the profit from each policy from the point of view of the insurance company.)

a) What is the probability distribution for x?
b) Calculate the expected value for x and explain what it means in the context of the problem?

141. A technician services mailing machines at companies on the Phoenix area. Depending on the type of malfunction, the service call can take 1, 2, 3, or 4 hours. The different types of malfunctions occur at about the same frequency.

a) Develop a probability distribution for the duration of a service call
b) Draw a graph of the probability distribution
c) Show that your probability distribution satisfies the conditions for a discrete probability function
d) What is the probability that a service call will take 3 hours?
e) A service call has just coming in and the type of malfunction is unknown. It is currently 3:00pm. Service technicians generally get off at 5pm. If they work longer they are paid overtime. What is the probability the service technician will have to work overtime to fix the machine today?

142. Spell-checking software catches “nonword” errors that result in a string of letters that is not a word, as when “the” is typed “teh". When undergraduates are asked to type a 250-word essay (without spell checking), the number X of non-word errors has the following distribution.

Value Of X	0	1	2	3	4
Probability	0.1	0.2	0.3	0.3	0.1

a) Draw a graph of the probability distribution of X
b) Write out the event that at least one nonword error occurs. What is the probability of this event?
c) What is the expected value of this probability distribution? Interpret this.
d) Describe the event X <= 2 in words. What is its probability?
e) What is the probability that X < 2?

Sampling (Midterm 2)

Remember that the **central limit theorem** states that the sampling distribution of a sample mean will be approximately Normal, no matter what population the sample is taken from, provided the sample size is large enough and the sample is a simple random sample. In this case, the sampling distribution of the sample mean is approximately normal with mean u and standard deviation (sigma / sqrt(n)).

Note that a **parameter** is a number that describes a population. In practice, the value of a parameter is not known because we can rarely examine the entire population.

On the other hand, a **statistic** is a number that can be computer from the sample data without making use of any unknown parameters. In practice, we often use a statistics to estimate an unknown parameter.

The **law of large numbers** states that, if we draw observations at random from any population with finite mean u, as the number of observations increases, the sample mean of the observed values tends to get closer and closer to the mean u of the population.

143. The time (in hours) that a technician requires to perform preventative maintenance on air conditioning is governed by the exponential curve. In this case, the mean time is 1 hour and the standard deviation is 1 hour. Imagine that the company has a contract to maintain 70 of these units in an apartment building.
    a) What is the probability that the average maintenance time exceeds 1.1 hours?
    b) What is the probability that the average maintenance time exceeds 1.25 hours?

144. Almost all medical schools in the United States require students to take the MCAT. To estimate the mean score u of those who took the test, you obtain the scores of simple random sample of students. Suppose that the scores follow a normal distribution, and from published information you know that the standard deviation is 6.5. Suppose that, unknown to you, the mean score of those taking the MCAT on your campus is 25.0.

    a) If you choose one student at random, what is the probability that the student's score is between 20 and 30?
    b) You sample 25 students. What is the sample distribution of their average score x?
    c) What is the probability that the mean score of your sample will be between 20 and 30?

145. On average, shoppers at Costco spend $61 and the standard deviation is $14. Suppose we take a simple random sample of 38 shoppers.

    a) Calculate the standard deviation of the sample mean
    b) Interpret the standard deviation of the sample mean?
    c) What is the probability that the sample mean will exceed $57?

146. According to reports, the average starting salary for college graduates is $34,500. (The standard deviation is known to be $3000). Let us assume that we take a simple random sample of 40 newly-employed recent graduates.

    a) What is the probability their the average salary is within $700 of the population average?
    b) In calculating the above probability, do we need to assume that the distribution of starting salary follows the normal distribution? Explain why or why not.
    c) If our sample size decreased, would the probability be larger or smaller than the one calculated above? Why?

147. Light vehicles sold in the United States must emit an average of no more than 0.05 grams per mile of nitrogen oxides after 50,000 or fewer miles of driving. NOX emissions after 50,000 or fewer miles of driving for one model car vary Normally with a mean of 0.03 and a standard deviation of 0.01.

a) What is the probability that a single car of this model emits more than 0.05 g/mi of NOX?
b) A company has 25 cars of this model in its fleet. What is the probability that the average NOX level (sample mean) of these cars is above 0.05 g/mi?

148. In a study of exercise, a large group of male runners walk on a treadmill for 6 minutes. After this exercise, their heart rates vary with mean 8.8 beats per 5 seconds and standard deviation of 1.0 beats per 5 seconds. The distribution takes only whole number values and is not normal.

a) Let x_bar be the mean number of beats per 5 seconds after measuring heart rate for 24 five-second intervals (2 minutes). What is the approximate distribution of x_bar (the sample mean) according to the central limit theorem?
b) What is the approximate probability that the sample mean is less than 8 beats per minute?
c) What is the approximate probability that the heart rate of a runner is less than 100 beats per minute? (Hint: Restate this event in terms of the sample mean x_bar)

Binomial Events & Binomial Probability (Midterm 2)

What is a binomial experiment?

- Experiment consists of a sequence of n identical trials
- Two outcomes are possible on each trial. We refer to one outcome as a success and one outcome as a failure.
- The probability of success, denoted by p, does not change from trial to trial. Consequently, the probability of a failure, denoted by 1-p, does not change from trial to trial.
- The trials are independent!

In a binomial experiment, our interest is in the number of successes occurring in the n trials! To solve these problems, we simply need to look at the experimental outcomes!

Eg. Suppose we want to count the number of heads occurring after flipping a coin 5 times!

a) Is this a binomial experiment?
    a. What is the random variable of interest? (Let x be the number of heads)
    b. What are the potential experimental outcomes? (0, 1, ….. 5)

Eg. Let's be marketers! Suppose that randomly call 3 people in a market research survey! And we know that the probability that any one customer on our call will make a purchase is 0.3.

a) Is this a binomial experiment?
    a. What is the random variable of interest? (Let x be the _____)
    b. What are the potential experimental outcomes for each trial?
    c. What are the total outcomes possible?
    d. Let's draw out the tree below of the possible options. Let S = Purchase and F = No Purchase. x = the total customers making a purchase.

What are the number of experimental outcomes resulting in exactly x successes in n trials? From your formula sheet:

	General Discrete Distribution	Binomial Distribution
Probability Equation	--	$f(x) = \frac{n!}{x! \cdot (n-x)!} p^x (1-p)^{n-x}$
Expected Value	$\mu = E(X) = \sum x \cdot f(x)$	$\mu = E(X) = np$

Note that the variance for a binomial random variable can be calculated as np(1-p). The number of experimental outcomes resulting in exactly x successes in n trials can be computed this way:

$$\frac{n!}{x! \cdot (n-x)!}$$

Note that n! = n(n-1)(n-2) …(2)(1), and that 0! = 1

Back to our example. Another question:

a) How many experimental outcomes result in two purchases?

$$\binom{n}{x} = \binom{3}{2} = \frac{3!}{2!\,(3-2)!} = \frac{(3)(2)(1)}{(2)(1)(1)} = \frac{6}{2} = 3$$

b) How many experimental outcomes result in three two purchases?

$$\binom{n}{x} = \binom{3}{3} = \frac{3!}{3!\,(3-3)!} = \frac{(3)(2)(1)}{3(2)(1)(1)} = \frac{6}{6} = 1$$

c) How do we find the probabilities? We have to add the probabilities? So what is the probability of the first two purchasing and the third not? (pp(1-p)) ….But we also have two other cases! So we need to simplify!

We need a general binomial probability function!

$$f(x) = \frac{n!}{x! \cdot (n-x)!} p^x (1-p)^{n-x}$$

Now, let's make a probability distribution! What are all the experimental outcomes? (0,1,2,3)

a) Probability of no purchases (x = 0)

$$\binom{3}{0} 0.3^0 (1-0.3)^{3-0} = 0.343$$

b) Probability of 1 purchase? (x = 1)
c) Probability of 2 purchases? (x = 2)
d) Probability of 3 purchases? (x = 3)

For each of these questions above, write out the final discrete probability distribution! (x & f(X)), Draw the graph of the final distribution, and calculate the expected value and variance of the distribution!

149. In a recent survey, teenagers and adults were asked how each group has listened to music in the past 12 months. A large majority of U.S. teenagers under the age of 18 say that they used Google's video sharing site YouTube, and **35%** of the teenagers in the study said that they used Spotify. Suppose that 10 teenagers are selected at random to ask how they listen to music.
   a) Is this a binomial experiment?
   b) What is the probability that none of the teenagers used Spotify?
   c) What is the probability that 4 out of the 10 teenagers used Spotify?
   d) What is the probability that at least 2 out of 10 teenagers use Spotify?
   e) What is the expected value for this problem? What is the variance?

150. The center for medicare reported that there were 295,000 appeals for hospitalization and other services. For this group, 40% of first-round appeals were successful. Suppose that 15 appeals have just been received.
   a) What is the probability that none of the appeals will be successful?
   b) What is the probability that exactly one of the appeals will be successful?
   c) What is the probability that at least two of the appeals will be successful?
   d) What is the expected value? What is the variance?

151. Let us say that you flip a coin 5 times! You are interested in the event of rolling a head (this = success).
   a) What are all the experimental outcomes?
   b) What is the probability of flipping 5 heads?
   c) What is the expected value? What is the variance of this question?

152. When a new machine is functioning properly, only 3% of the items produced are defective. Assume that we will randomly select two parts produced on the machine and that we are interested in the number of defective parts found.
   a) Describe the conditions under which the situation would be a binomial experiment?
   b) How many experimental outcomes result in exactly one defect being found?
   c) Compute the probabilities associated with finding no defects, exactly one defect, and two defects?

153. According to a 2017 study by the Pew Research Center, 15% of adults in the US do not use the internet. Suppose that 10 adults in the US are selected randomly.
   a. Is the selection of 10 adults a binomial experiment? Explain.
   b. What is the probability that none of the adults use the internet?
   c. What is the probability that 3 of the adults use the internet?
   d. What is the probability that at least 1 of the adults uses the internet?

154. Military radar and missile detection systems are designed to warn a country of an enemy attack. A reliability question is whether a detection system will be able to identify an attack and issue a warning. Assume that a particular system has a 0.9 probability of detecting a missile attack. Use a binomial probability distribution to answer the following questions.

a. What is the probability that a single detection system will detect and attack?
b. If two detection systems are installed in the same area and operate independently, what is the probability that at least one of the systems will detect an attack?
c. If three systems are installed, what is the probability that at least one of the systems will detect the attack?
d. Would you recommend that multiple systems be used? Explain?

155. Market-share analysis monitors and reports on Internet browser usage. According to this analysis, during the summer of 2019, Google Chrome's browser exceeded 30% market share with a 30.37% share of the browser market. Assume that we randomly select a group of 20 internet browsers. Answer the following questions.
a. Compute the probability that 8 out of 20 internet browser users use Chrome as their internet browser
b. Compute the probability that at least 3 of the 20 internet browser users use Chrome as their internet browser.
c. For the sample of 20 internet browsers, computer the expected number of chrome users as well as the variance and standard deviation for the number of chrome users.

156. A study conducted by Pew Research center showed that 75% of 18 to 34 year olds living with their parents say that they contribute to household expenses. Suppose that a random sample of fifteen, 18 to 34 year olds living with their parents is selected and ask whether they contribute to household expenses.
a. Is the selection of the fifteen 18 to 34 year olds living with their parents a binomial experiment. Explain?
b. If the sample shows that none of the fifteen 18 to 34 year olds living with their parents contributes to household expenses, would you question the results of the study? Why and how?
c. What is the probability that at least 10 of the 15 18 to 34 year olds living with their parents contribute to household expenses?

157. A university found that 20% of its students withdraw without completing the introductory statistics course.
a. Compute the probability that 2 or fewer will withdraw?
b. Compute the probability that exactly 4 will withdraw?
c. Compute the probability that more than 3 will withdraw?
d. Compute the expected number of withdrawals as well as the variance and standard deviation?

158. A recent poll showed that 30% of Canadians were satisfied with the way that things are going in Canada. Suppose a sample of 20 Canadians is selected as part of a study of the state of the nation.
a. Compute the probability that exactly 4 of the 20 Canadians surveyed are satisfied with the say things are going in Canada?
b. Compute the probability that at least 2 of the Canadians surveyed are satisfied with the way things are going in Canada?
c. For the sample of 20 Canadians, compute the expected number of Canadians who are satisfied with the way things are going in Canada? What is the variance and standard deviation?

159. According to a recent study on social media analytics, 71% of all tweets get no reaction. That is, these tweets are not replied to or retweeted. Suppose that we select 100 tweets.
a. What is the expected number of these tweets with no reaction? What is the variance and standard deviation for these tweets?

160. PBS News Hour recently reported that 39.4% of Americans between the ages of 25 and 64 have at least a two-year college degree. Assume that 50 Americans between the ages between the ages of 25 and 64 are randomly selected.
    a. What is the expected number of people with at least a two year college degree?
    b. What is the variance and standard deviation for the number of people with at least a two-year college degree?

161. Mahoney Custom Home Builders, Inc. of Canyon Lake, Texas, asked visitors to their website what is the most important when choosing a home builder. Possible responses were quality, price, customer referral, years in business, and special features. Results showed that 23.5% of the respondents chose price as the most important factor. Suppose a sample of 200 potential home buyers in the Canyon Lake area were selected.
    a. How many people would you expect to choose price at the most important factor when choosing a home builder?
    b. What is the standard deviation of the number of respondents who would choose price as the most important factor in selecting a home builder.
    c. What is the standard deviation of the number of respondents who do not list price as the most important factor in selecting a home builder.

162. It is estimated that 80% of car buyers purchase small cars. Suppose fifteen randomly selected individuals in a spa say they intend to purchase a vehicle within the next three months; assume that the purchase decision of these individuals is a binomial experiment.
    a. What are the n, x, and p for this problem, assuming that we treat it as a binomial experiment?
    b. What is the chance that exactly twelve individuals in the sample will purchase small cars within the next three months?

Hypothesis Testing (Final Exam)

163. The Federal Trade Commission (FTC) periodically tests claims that manufacturers make about their products. Let us imagine that a certain coffee producer claims that the can contains 3 pounds of coffee per can. This is the mean. In this case, success means that the amount of coffee in the can must be 3 or more pounds to satisfy the FTC regulations. Let us assume that we sample 36 cans of coffee and find that the average amount of coffee in the cans is 2.92, with a standard deviation (population) assumed to be 0.18 ounces.
    a. What type of hypothesis test will you conduct in this case?
    b. What are the null and alternative hypothesis for this problem?
    c. Compute the relevant test statistic for this test and determine the p value for this test?
    d. Review the results from this test, reject or fail to reject the null hypothesis.

164. Previous studies have concluded that home owners live in their homes for an average of 8.4 years before selling them. A realty company wishes to investigate if this figure has changed. A sample of 60 families who have recently sold their house was selected. The mean time that these families had lived in their homes was 8.94 years. The standard deviation was 3.49 years.
    a. What can be concluded from these data?
    b. Test the appropriate hypotheses at alpha = .05. Report the p-value.

165. Do college students work too much? According to a 2015 report, 70% of college students in Canada have a full or part time job while enrolled in college. An administrator from a local college questions the accuracy of this claim. In particular, she believes that the true proportion of college students who have a full or part time job while studying at her college is different from 70%. The administers a SRS and finds that 238 of the 325 students have a part time job.
    a. What type of hypothesis test will you conduct?

b. What is the null hypothesis for this problem? The alternative hypothesis?
c. What is the p value obtained for the data in question?
d. Can you accept or reject the data for this question?

Producing Data (Sampling) (Midterm 2)

A **population** in a statistical study is the entire group of individuals about which we want information.

A **sample** is a part of the population from which we actually collect information. We use a sample to draw conclusions about the entire population.

A **sampling design** describes exactly how we choose to sample from the population.

The design of a statistical sample is **biased** if it systematically favors certain outcomes.

A **voluntary response sample** consists of people who choose themselves by responding to a broad appeal. Voluntary response samples are biased because people with strong opinions are more likely to respond.

A **simple random sample** (SRS) of size n consists of n individuals from the population chosen in such at way that every set of n individuals has an equal chance to be the sample actually selected.

A **stratified random sample** first classifies a population into groups of similar individuals, called strata. Then, the sampler can choose a separate SRS in each stratum and combine those to form the sample. This might be completed if you believe that different groups will have very different responses to each other.

Two key types of bias that are present when sampling are **response bias** and **wording effects**. In response bias, individuals may provide a response that they think will be favourably received by the interviewer. On the other hand, wording effects relate to the fact that confusing or leading questions can introduce strong bias and variations in wording that can change outcomes in a survey.

An **observational study** observes individuals and measures variables of interest but does not attempt to influence the responses. The purposes of an observational study is to describe some group or situation.

An **experiment** deliberately imposes some treatment on individuals to observe their responses. The purpose of an experiment is to study whether the treatment causes a change in the response.

A **lurking** variable is a variable that is not among the explanatory or response variables in a study and yet may influence the interpretation of relationships among those variables.

166. A political scientist wants to know how college students feel about the social security system. She obtains a list of the 3456 undergraduates at her college and mails a questionnaire to 250 students selected at random. Only 104 questionnaires are returned.

a) What is the population in this study? Be careful: What group does she want information about?
b) What is the sample? Be careful: From what group does she actually obtain information?

167. An archeological dig turns up large numbers of pottery shards, broken stone implements, and artifacts. Students working on the project classify each artifact and assign it a number. The counts in different categories are important for understanding the site, so the project director chooses 2% of the artifacts at random and checks the students work

a) What are the population and sample here?

168. Many observational studies shows that people who drink a moderate amount of alcohol have less heart disease than people who drink no alcohol or who drink heavily? (Moderate means one or two drinks a day for men and one drink a day for women)

a) What are the explanatory and response variables?
b) Moderate drinkers as a group tend to be more likely to maintain a healthy weight, get enough sleep, and exercise regularly than those who drink no alcohol or drink heavily. Are the variables maintaining a healthy weight, getting enough sleep, and exercising regularly explanatory variables, response variables, or lurking variables? Explain your reason?
c) Is the association between drinking and heart disease good reason to think that moderate drinking actually causes less heart disease? Explain why or why not?

Tables & Formulas

The Chi-square(χ^2)test:

- Expected count for a cell in a two-way table:

$$expected\ count = \frac{row\ total * column\ total}{table\ total}$$

- Chi-square test statistic for testing whether the row and column variables in an r x c table are unrelated (expected cell counts not too small):

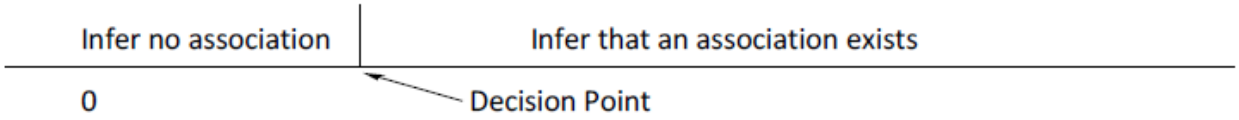
$$\chi^2 = \sum \frac{(observed\ count - expected\ count)^2}{expected\ count}$$

With decision points from the Chi-square distribution with df=(r-1)x(c-1)

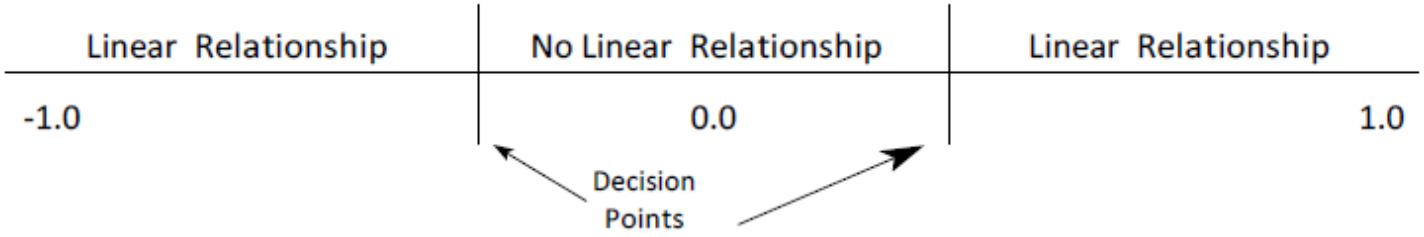
Chi-square decision points for various degrees of freedom (d.f.)

d.f. = (# of rows - 1)(# of columns - 1) = (r - 1)(c - 1)

D.F.	Decision Point	D.F.	Decision Point	D.F.	Decision Point
1	3.84	14	23.68	2	38.89
2	5.99	15	25.00	27	40.11
3	7.81	16	26.30	28	41.34
4	9.49	17	27.59	29	42.56
5	11.07	18	28.87	30	43.77
6	12.59	19	30.14	40	55.8
7	14.07	20	31.41	50	67.5
8	15.51	21	32.67	60	79.1
9	16.92	22	33.92	70	90.5
10	18.31	23	35.17	80	102.0
11	19.68	24	36.42	90	113.0
12	21.03	25	37.65	100	124.3
13	22.36				



Decision Points for determining the significance of linear correlation coefficient, r, from a random sample of size n:



n	Decision Point	n	Decision Point	n	Decision Point
5	0.878	14	0.532	26	0.388
6	0.811	15	0.514	28	0.374
7	0.754	16	0.497	30	0.361
8	0.707	17	0.482	40	0.312
9	0.666	18	0.468	50	0.279
10	0.632	19	0.456	60	0.254
11	0.602	20	0.444	80	0.220
12	0.576	22	0.423	100	0.196
13	0.553	24	0.404	200	0.139

Exploring Data: Relationships

- Look for overall pattern (form, direction, strength) and deviations (outliers, influential observations).
- Correlation coefficient, r:

$$r = \frac{1}{n-1}\sum_{i=i}^{n}\left(\frac{x_i - \bar{x}}{s_x}\right)\left(\frac{y_i - \bar{y}}{s_y}\right)$$

- Least-squares regression line:

$$\hat{y} = a + bx$$

With slope $b = r\frac{s_y}{s_x}$ and intercept $a = \bar{y} - b\bar{x}$

- Residuals: $e = y - \hat{y}$ (residual=observed y – predicted y)

- Prediction interval:

$$\hat{y} \pm z * RMS_{error}$$

Table entry for z is the area under the standard normal curve to the left of z.

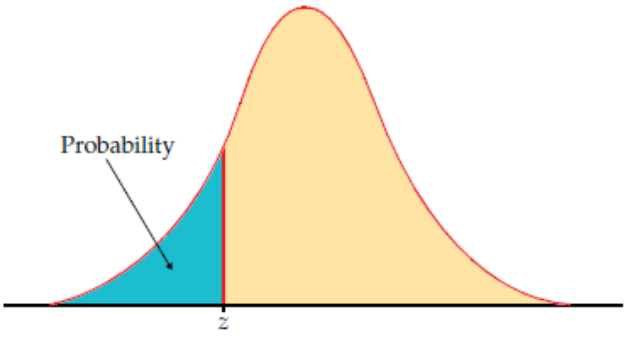


TABLE A

Standard normal probabilities

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
−3.4	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0002
−3.3	.0005	.0005	.0005	.0004	.0004	.0004	.0004	.0004	.0004	.0003
−3.2	.0007	.0007	.0006	.0006	.0006	.0006	.0006	.0005	.0005	.0005
−3.1	.0010	.0009	.0009	.0009	.0008	.0008	.0008	.0008	.0007	.0007
−3.0	.0013	.0013	.0013	.0012	.0012	.0011	.0011	.0011	.0010	.0010
−2.9	.0019	.0018	.0018	.0017	.0016	.0016	.0015	.0015	.0014	.0014
−2.8	.0026	.0025	.0024	.0023	.0023	.0022	.0021	.0021	.0020	.0019
−2.7	.0035	.0034	.0033	.0032	.0031	.0030	.0029	.0028	.0027	.0026
−2.6	.0047	.0045	.0044	.0043	.0041	.0040	.0039	.0038	.0037	.0036
−2.5	.0062	.0060	.0059	.0057	.0055	.0054	.0052	.0051	.0049	.0048
−2.4	.0082	.0080	.0078	.0075	.0073	.0071	.0069	.0068	.0066	.0064
−2.3	.0107	.0104	.0102	.0099	.0096	.0094	.0091	.0089	.0087	.0084
−2.2	.0139	.0136	.0132	.0129	.0125	.0122	.0119	.0116	.0113	.0110
−2.1	.0179	.0174	.0170	.0166	.0162	.0158	.0154	.0150	.0146	.0143
−2.0	.0228	.0222	.0217	.0212	.0207	.0202	.0197	.0192	.0188	.0183
−1.9	.0287	.0281	.0274	.0268	.0262	.0256	.0250	.0244	.0239	.0233
−1.8	.0359	.0351	.0344	.0336	.0329	.0322	.0314	.0307	.0301	.0294
−1.7	.0446	.0436	.0427	.0418	.0409	.0401	.0392	.0384	.0375	.0367
−1.6	.0548	.0537	.0526	.0516	.0505	.0495	.0485	.0475	.0465	.0455
−1.5	.0668	.0655	.0643	.0630	.0618	.0606	.0594	.0582	.0571	.0559
−1.4	.0808	.0793	.0778	.0764	.0749	.0735	.0721	.0708	.0694	.0681
−1.3	.0968	.0951	.0934	.0918	.0901	.0885	.0869	.0853	.0838	.0823
−1.2	.1151	.1131	.1112	.1093	.1075	.1056	.1038	.1020	.1003	.0985
−1.1	.1357	.1335	.1314	.1292	.1271	.1251	.1230	.1210	.1190	.1170
−1.0	.1587	.1562	.1539	.1515	.1492	.1469	.1446	.1423	.1401	.1379
−0.9	.1841	.1814	.1788	.1762	.1736	.1711	.1685	.1660	.1635	.1611
−0.8	.2119	.2090	.2061	.2033	.2005	.1977	.1949	.1922	.1894	.1867
−0.7	.2420	.2389	.2358	.2327	.2296	.2266	.2236	.2206	.2177	.2148
−0.6	.2743	.2709	.2676	.2643	.2611	.2578	.2546	.2514	.2483	.2451
−0.5	.3085	.3050	.3015	.2981	.2946	.2912	.2877	.2843	.2810	.2776
−0.4	.3446	.3409	.3372	.3336	.3300	.3264	.3228	.3192	.3156	.3121
−0.3	.3821	.3783	.3745	.3707	.3669	.3632	.3594	.3557	.3520	.3483
−0.2	.4207	.4168	.4129	.4090	.4052	.4013	.3974	.3936	.3897	.3859
−0.1	.4602	.4562	.4522	.4483	.4443	.4404	.4364	.4325	.4286	.4247
0.0	.5000	.4960	.4920	.4880	.4840	.4801	.4761	.4721	.4681	.4641

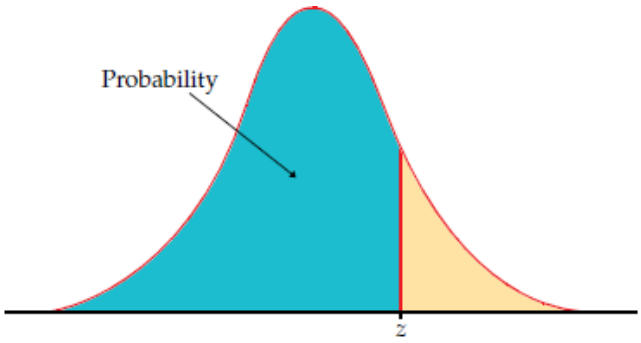


Table entry for z is the area under the standard normal curve to the left of z.

TABLE A

Standard normal probabilities (continued)

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	.5000	.5040	.5080	.5120	.5160	.5199	.5239	.5279	.5319	.5359
0.1	.5398	.5438	.5478	.5517	.5557	.5596	.5636	.5675	.5714	.5753
0.2	.5793	.5832	.5871	.5910	.5948	.5987	.6026	.6064	.6103	.6141
0.3	.6179	.6217	.6255	.6293	.6331	.6368	.6406	.6443	.6480	.6517
0.4	.6554	.6591	.6628	.6664	.6700	.6736	.6772	.6808	.6844	.6879
0.5	.6915	.6950	.6985	.7019	.7054	.7088	.7123	.7157	.7190	.7224
0.6	.7257	.7291	.7324	.7357	.7389	.7422	.7454	.7486	.7517	.7549
0.7	.7580	.7611	.7642	.7673	.7704	.7734	.7764	.7794	.7823	.7852
0.8	.7881	.7910	.7939	.7967	.7995	.8023	.8051	.8078	.8106	.8133
0.9	.8159	.8186	.8212	.8238	.8264	.8289	.8315	.8340	.8365	.8389
1.0	.8413	.8438	.8461	.8485	.8508	.8531	.8554	.8577	.8599	.8621
1.1	.8643	.8665	.8686	.8708	.8729	.8749	.8770	.8790	.8810	.8830
1.2	.8849	.8869	.8888	.8907	.8925	.8944	.8962	.8980	.8997	.9015
1.3	.9032	.9049	.9066	.9082	.9099	.9115	.9131	.9147	.9162	.9177
1.4	.9192	.9207	.9222	.9236	.9251	.9265	.9279	.9292	.9306	.9319
1.5	.9332	.9345	.9357	.9370	.9382	.9394	.9406	.9418	.9429	.9441
1.6	.9452	.9463	.9474	.9484	.9495	.9505	.9515	.9525	.9535	.9545
1.7	.9554	.9564	.9573	.9582	.9591	.9599	.9608	.9616	.9625	.9633
1.8	.9641	.9649	.9656	.9664	.9671	.9678	.9686	.9693	.9699	.9706
1.9	.9713	.9719	.9726	.9732	.9738	.9744	.9750	.9756	.9761	.9767
2.0	.9772	.9778	.9783	.9788	.9793	.9798	.9803	.9808	.9812	.9817
2.1	.9821	.9826	.9830	.9834	.9838	.9842	.9846	.9850	.9854	.9857
2.2	.9861	.9864	.9868	.9871	.9875	.9878	.9881	.9884	.9887	.9890
2.3	.9893	.9896	.9898	.9901	.9904	.9906	.9909	.9911	.9913	.9916
2.4	.9918	.9920	.9922	.9925	.9927	.9929	.9931	.9932	.9934	.9936
2.5	.9938	.9940	.9941	.9943	.9945	.9946	.9948	.9949	.9951	.9952
2.6	.9953	.9955	.9956	.9957	.9959	.9960	.9961	.9962	.9963	.9964
2.7	.9965	.9966	.9967	.9968	.9969	.9970	.9971	.9972	.9973	.9974
2.8	.9974	.9975	.9976	.9977	.9977	.9978	.9979	.9979	.9980	.9981
2.9	.9981	.9982	.9982	.9983	.9984	.9984	.9985	.9985	.9986	.9986
3.0	.9987	.9987	.9987	.9988	.9988	.9989	.9989	.9989	.9990	.9990
3.1	.9990	.9991	.9991	.9991	.9992	.9992	.9992	.9992	.9993	.9993
3.2	.9993	.9993	.9994	.9994	.9994	.9994	.9994	.9995	.9995	.9995
3.3	.9995	.9995	.9995	.9996	.9996	.9996	.9996	.9996	.9996	.9997
3.4	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9998